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Bioactive proteins in bovine colostrum and effects of heating, drying and irradiation

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Running title: Processing affects bioactive proteins in colostrum

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Abstract

Bovine colostrum (BC) contains bioactive proteins, such as immunoglobulin G (IgG), lactoferrin (LF) and lactoperoxidase (LP). BC was subjected to low-temperature, long-time pasteurization (LTLT, 63 °C, 30 min) or high-temperature, short-time pasteurization (HTST, 72 °C, 15 s) and spray-drying (SD), with or without γ-irradiation (GI, ~14 kGy) to remove microbial contamination. Relative to unpasteurized liquid BC, SD plus GI increased protein denaturation by 6 and 11%, respectively, increasing to 19 and 27% after LTLT and to 48% after HTST, with no further effects after GI (all $P < 0.05$). LTLT, without or with GI, resulted in 15 or 29% denaturation of IgG, compared with non-pasteurized BC, and 34 or 58% for HTST treatment (all $P < 0.05$, except LTLT without GI). For IgG, only GI, not SD or LTLT, increased denaturation (30-38%, $P < 0.05$) but HTST increased denaturation to 40%, with further increases after GI (60%, $P < 0.05$). LTLT and HTST reduced LP levels (56 and 81% respectively) and LTLT reduced LF levels (21%), especially together with GI (47%, $P < 0.05$). Denaturation of BSA, β-LgA, β-LgB and α-La were similar to IgG. Methionine, a protective amino acid against free oxygen radicals, was oxidised by LTLT+GI ($P < 0.05$) while LTLT and HTST had no effect. Many anti-inflammatory proteins, including serpin anti-proteinases were highly sensitive to HTST and GI but preserved after LTLT pasteurization. LTLT, followed by SD is an optimal processing technique preserving bioactive proteins when powdered BC is used as a diet supplement for sensitive patients.

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Introduction

Human milk is rich in bioactive proteins, including immunoglobulins (IG), lactoferrin (LF) and lactoperoxidase (LP), milk fat globule membrane proteins, osteopontin and lysozyme. [1] LP participates in antimicrobial defence by catalysing the oxidation of thiocyanate ions in the presence of hydrogen peroxide, thereby forming protective hypothiocyanate ions, [2, 3] whereas LF has antibacterial and anti-inflammatory activity, in particular in the intestine. [4] Infant formula (IF) is predominantly based on bovine milk and while the levels of macronutrients are manipulated to mimic those in human milk, the levels of bioactive proteins are much lower. [5] Several thermal treatments are used in formula production (e.g. pasteurization, ultra-high-temperature processing, UHT or spray-drying, SD) which may induce protein denaturation and lower bioactivity. [6] UHT-treatment applied to milk or high-lactose-containing IFs, rich in unsaturated fatty acids and iron, induces Maillard reactions which followed by long storage of the product leads to lipid and protein oxidation and aggregation. These effects can influence the shelf life of a formula[7] and lower the amounts of native proteins and induce the formation of advanced glycosylation end products (AGE). Following feeding, Maillard-modified proteins have been proposed to bind to AGE receptors and thus activate NF-κB and induce inflammation. [8]

It is well known that IF is suboptimal in protecting newborn infants against serious immune-and microbiota-related disorders, such as necrotizing enterocolitis (NEC). On the other hand, bovine colostrum (BC) which is secreted from the mammary gland during the first 1-2 days postpartum, contains high levels of bioactive proteins, including increased levels of IG, LF and LP, bovine serum albumin (BSA), growth factors, α-lactalbumin (α-La) and β-lactoglobulin (β-Lg). [1, 9, 10] Many of these BC proteins may help to ensure the optimal growth and development of the intestinal and immune system in infants, when mother´s own milk is not available. BC also provides

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a very high total antioxidant capacity for protection against oxidative stress in these infants. [11] IgG is present in BC as two forms, IgG1 and IgG2, the concentration of IgG1 being approximately 10 times greater than that of IgG2. [12] However, β-lactoglobulin (β-Lg), is a protein which is absent in human colostrum and human term milk and is involved in aggregation reactions with other proteins through its free thiol group which becomes exposed during heating. [1, 13] Studies using preterm pigs have demonstrated a reduced incidence and severity of NEC and diarrhoea following feeding of untreated, heat-treated or spray-dried BC, compared with IF. [14, 15] A powdered BC product, subjected to pasteurization at 63 °C for 30 min, mild SD and γ-irradiation (GI) has been tested in two pilot studies in preterm infants, as a supplement during the first 10 days after life with no adverse events[16], and larger international trials are now underway. [17] Further, BC supplementation to children subjected to cancer therapy has been tested with beneficial effects. [18] It remains unclear if the processing methods used are critical for the bioactivity of the BC product in these trials and there is limited information on the effects of heat treatment on BC or human colostrum bioactive proteins.

Pasteurization of human donor milk (DM) is performed either by low temperature/ long time (LTLT) pasteurization at 63 °C for 30 min or by standard high temperature/ short time pasteurization (HTST, 72 °C for 15 sec). [19, 20] During LTLT pasteurization 10-16% aggregation of IgG occurs which affects its biological activity by modulating Fcγ receptor binding. [21] LTLT pasteurization also lowers the general nutritional value of DM provided to neonates. [6, 22] In other studies, LTLT pasteurization lowered IgG levels in human milk by 34%. [23] To further reduce the risk of bacterial contamination during packaging, products for such sensitive newborns can be subjected to GI, a non-thermal method to improve microbial safety via generation of free radicals and reactive oxygen species. Irradiation of IF has been used to prevent the growth of C. sakazakii and Salmonella in powdered milk IF. [24-26] It appears that GI at 10 kGy dose has no major effects on

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80 milk proteins, carbohydrates and fats, with some effects occurring at 50 kGy, partly due to changes
81 in amino acid composition induced by oxidative losses. [27] Thus, the industrial processing conditions
82 (LTLT, HTST, SD, GI) may affect the bioactivity of BC when used for highly sensitive patient
83 groups, such as preterm infants. We hypothesized that the variable intensity of BC processing
84 (pasteurization temperature and time), with or without gentle nozzle and continuous flow-through
85 belt SD, conventional SD and GI (14 kGy), would affect the denaturation and aggregation of key
86 bioactive proteins in powdered BC products and that methionine oxidation may be a useful marker
87 of oxidative effects occurring during processing. To determine this, we applied a novel combination
88 of BC processing conditions (including pasteurisation and spray drying with and without GI) and
89 analytical technologies (protein separation and aggregation, western blot and proteomics analyses)
90 to study the thermal and oxidative stability of key bioactive proteins in BC and to identify
91 protective and anti-inflammatory proteins which could be beneficial for these compromised infants.

92 Materials and methods

93 Protein contents and nitrogen solubility index

94 BC was isolated from Danish dairy cows and immediately stored frozen. Prior to processing, forzen
95 BC was gently thawed overnight in a bulk tank. In all, two liquid BC samples and six BC powders
96 were produced under different processing conditions, from the same batch of raw liquid colostrum,
97 by a commercial company in Denmark (Biofiber Damino, Gesten, Denmark, Table 1). Processing
98 included gentle LTLT pasteurization and FILTERMAT drying, or HTST pasteurization and
99 conventional SD. Samples were either not subjected to GI or alternatively treated with GI (14 kGy,
100 Sterigenics, Belgium) after SD. Control samples were either raw liquid BC (RBC) or unpasteurized,
101 spray-dried BC without GI (BC00-γ). The powders were mixed for 1 hour on a magnetic stirrer,
102 with Milli-Q water to a 1% (wt/vol) protein solution and left at 4 °C, overnight to ensure protein

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hydration. Nitrogen content of all samples was determined via the Kjeldahl method (IDF, 1993) and multiplied by the conversion factor of 6.38, to calculate the protein content. The analysis was performed in duplicate or triplicate with a positive control and blanks in duplicate. The nitrogen content was calculated through the following equation:

$$N = \frac{1.40067 * (Vol_{HCl} - Vol_{Blank}) * M_{HCl}}{m_{sample}}$$

For nitrogen solubility index (NSI), the procedure of Carr (1999) was followed with modifications. Samples were reconstituted to a protein content of 3.5% (w/v). 40 ml Milli-Q water was added to 10 ml of the reconstituted sample and 25 ml of this solution was adjusted to pH 4.6 with 0.1 M HCl. After precipitation for 30 min the samples were filtered through a 3W filter paper (Ahlstrom Munktell, Bärenstein, Germany) and 10 ml of the filtrate was used for Kjeldahl analysis, with 10 ml filtrate replacing 1.0 g powder in the original method. Samples were titrated with 0.1 M HCl. The percentage denaturation was calculated relative to the control, RBC 0h, as follows:

$$Denaturation\ [\%] = \frac{Protein_{control} - Protein_{sample}}{Protein_{control}} * 100\%$$

Protein separation and aggregate analyses

Sodium dodecyl sulphate polyacrylamide gel electrophoresis (SDS-PAGE) was performed on all samples. For the analysis of non-aggregated proteins, 1% (w/v) protein solutions were prepared and the supernatant was analysed after centrifugation at 10,000 x g for 15 min and removal of lipids. For initial protein identification, samples were analysed under reducing and non-reducing conditions and compared to reduced or non-reduced protein standards (α-La, β-Lg, BSA, LF, LP and IgG; Sigma Aldrich, St. Louis, USA). For reduction, 0.5 M DTT solution was added to the samples to a total concentration of 50 mM DTT and incubated at 56 °C for 30 min. Alkylation of the reduced samples was performed by addition of a 500 mM iodoacetamide solution to a

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125 concentration of 10 mM iodoacetamide followed by placement in the dark for 30 min before
126 performing SDS-PAGE. For both analyses, samples were diluted and standardized to the same
127 protein concentration (1% w/v) before addition of 4 X LDS sample buffer and heated at 80 °C for
128 10 min. 10 µl of each sample (corresponding to 7.7 µg of total protein) were loaded in each well of
129 a precast 12% Bis-Tris gel (NuPAGE, Thermo Fischer Scientific, Roskilde, Denmark). A broad
130 range molecular weight marker (Bio-Rad Laboratories, Hercules, CA, USA) was used for
131 comparison and proteins were separated at 200 V with NuPAGE MOPS SDS running buffer
132 (Thermo Fischer Scientific). Proteins were stained with 0.2% Coomassie Brilliant Blue R-250
133 solution in phosphoric acid, ammonium sulphate and 96% ethanol. The gels were destained in
134 Milli-Q water and scanned using an Epson Perfection V750 Pro scanner. Comparison between gel
135 bands was based on volumes computed by using Phoretix TL120 software, with rolling-ball 90
136 algorithm for background subtraction.

137 Western blotting was performed following SDS-PAGE under non-reducing conditions
138 as indicated previously. [28] Briefly, 0.1 or 5 µg of BC for IgG or LF respectively, was applied per
139 lane and compared to a prestained protein molecular weight marker (PageRuler, Thermo Fisher
140 Scientific). Proteins were resolved by SDS-PAGE and transferred onto PVDF membranes using an
141 iBlot2 system (Thermo Fisher Scientific). For IgG, proteins were probed with rabbit anti-bovine
142 IgG HRP-conjugated antibody (Thermo Fisher Scientific), diluted 1:3000 in TBST-BSA, or for LF,
143 by goat anti-bovine LF antibody (Bethyl Laboratories, Montgomery, Texas, USA) diluted 1:3000 in
144 TBST-BSA, followed by Donkey anti-goat IgG, HRP conjugate (Thermo Fisher Scientific) diluted
145 1:3000 in TBST-BSA. Proteins were detected by chemiluminescence using Clarity Western ECL
146 Substrate, Bio-Rad and captured by a ChemiDoc imaging system (BioRad, Hercules, CA, USA)
147 followed by quantification via ImageLab software. Total IgG was determined by a sandwich

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148 ELISA, using a sheep anti-bovine IgG (A10-118A-12) capture antibody and a HRP-conjugated
149 sheep anti-bovine IgG antibody (A10-118P-28, Bethyl Laboratories, Montgomery, Texas, USA).

150 Protein precipitation and trypsin hydrolysis

151 Proteins were precipitated by treatment with a mixture of 10% TCA, 80% Acetone in Milli-Q water
152 and incubated on ice for 2 h. Samples were mixed every 30 min for 2 h and thereafter centrifuged at
153 at 10,000 x g for 30 min at 4 °C. The resulting pellet was washed with 200 µl of 80% acetone and
154 incubated on ice for 15 min. After centrifugation at 10,000 x g for 30 min at 4 °C, the remaining
155 protein pellet was reconstituted in 12.5 µL of a solution consisting of 8 M urea, 50 mM Tris-HCl,
156 pH 8.0 and 2 µl of 450 mM DTT and incubated at room temperature for 45 min. The sample was
157 then alkylated by incubating with 4 µl of 500 mM iodoacetamide at room temperature for 1 h in the
158 dark. Sample aliquots containing ~100 µg of protein were diluted with ammonium bicarbonate
159 buffer (50 mmol/L) to 66 µL, and hydrolyzed with trypsin (Trypsin/Lys-C, LC-MS grade, Promega
160 Corporation, Madison, USA, 10 µl of 0,1 µg/µl), at 37 °C for 18 h. The hydrolysis was stopped by
161 addition of 5 µL of 20% TFA, thereafter samples were stored at -20 °C. Samples were analysed in
162 three technical replicates (i.e. the whole procedure was performed for three different aliquots of
163 each BC sample in parallel starting with protein reconstitution).

164 Proteomics analyses

165 The procedure was as described previously, with minor modifications. [29] Analysis was carried out
166 on an ultra HPLC (UHPLC) Ultimate 3000 (Thermo Fisher Scientific) with a C18 column (Aeris
167 peptide C18), with dimensions of 150 x 2.1 mm, 1.7 µm, 100 Å at 40°C (Phenomenex, Værløse,
168 Denmark). 10 µl sample was injected and a flow rate of 0.25 ml/min was used. The gradient was
169 100% of buffer A (0.1% formic acid in MilliQ water) for 5 min followed by a linear increase from
170 0-60% buffer B (0.1% formic acid in 80% acetonitrile) over 45 min. The HPLC was connected to a

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171 Q Exactive mass spectrometer (Thermo Fisher Scientific) for MS/MS analysis of the separated
172 peptides. For the analysis of the mass spectra, the software Thermo Proteome Discoverer 2.2.0.388
173 (Thermo Fisher Scientific) was used. Sequest HT was used to search against the Bos taurus
174 proteome in the Swissprot database (SwissProt TaxID=9913, v2017-04-12. Precursor mass
175 tolerance was set at 10 ppm, fragment mass tolerance at 0.6 Da; Lysine carboxymethylation
176 (+58.005 Da) and carboxyethylation (+72.021 Da), methionine oxidation (+15.995 Da) and hexose
177 glycation (+162.053 Da) were set as variable modifications and cysteine carbamidomethylation
178 (+57.021 Da) as fixed modification. Only modified peptides with high confidence levels were
179 selected in order to reduce the number of false positives. Additionally, the identified glycated
180 peptides were confirmed by visual inspection of each spectrum. All confirmed hexose-modified
181 peptides were combined in one timed inclusion list to target them for label-free quantification. The
182 average of the peak areas obtained for each peptide, analyzed in three technical replicates (three
183 separate samples from protein reconstitution) was normalized to the highest peak area calculated in
184 any of the BC samples (100%), i.e. relative peak areas are expressed on a relative scale from 0%
185 (below limit of quantification) to 100% (highest content).

186 **Statistical analysis**

187 Statistical analysis was performed using GraphPad Prism 8.3.0 (GraphPad Software, La Jolla, CA,
188 USA). Unless otherwise stated, ANOVA with Tukey's post-test and corrections for multiple
189 comparisons was used for comparison of results. Samples used for statistical analysis were
190 performed in duplicates of the technical triplicates.

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Results

Protein denaturation and total IgG in BC samples

During NSI analysis, denatured BC proteins associated with casein micelles following processing are precipitated at pH 4.6 and these proteins are removed by filtration. Thus, NSI analysis provides an indication of protein denaturation. Protein denaturation in all cases was expressed relative to raw unpasteurized liquid BC (RBC 0 h). Thus, relative to RBC 0 h, SD without prior pasteurization, nor GI (BC00-γ) resulted in an increased denaturation by 6% ($P < 0.05$). Likewise, relative to RBC 0 h, GI of the unpasteurised, SD colostrum powder (i.e. BC00+γ) further increased protein denaturation to 12%. LTLT pasteurization at 63 °C, 30 min without GI (BC63-γ) increased denaturation to 19% rising to 28% following GI of the SD powder (BC63+γ). HTST pasteurization at 72 °C, 15 sec and SD (BC72-γ) extended denaturation to 48%, but no differences were observed following pasteurization at 72 °C and SD and GI (BC72+γ, all $P < 0.05$, Fig. 1A).

The total antigenicity of IgG was measured by ELISA and any changes (loss or increase) in quantity reflect thermally-induced denaturation and modulation of antigenicity during processing. Total antigenicity of IgG in all cases was expressed relative to raw unpasteurized liquid BC (RBC 0 h), which had been subjected to SD but no GI (i.e. BC00-γ). Thus, compared to BC00-γ, total antigenic IgG decreased in accordance with the severity of processing. GI of non-pasteurized BC induced a loss of IgG antigenicity of 21% compared with the control (BC00-γ vs. BC00+γ), which was greater than that caused by LTLT pasteurization (15%, BC63-γ). LTLT plus GI resulted in a further reduction in IgG (29%). Compared to the effects of LTLT pasteurization, HTST pasteurization resulted in a further loss of IgG antigenicity to 32% and HTST + GI resulted in a loss of IgG of 59%. As protein aggregates form during heat-induced denaturation during processing, samples were also centrifuged (suffixed with "+C") to remove insoluble aggregates and the supernatant analysed (Fig. 1B). Only few differences in IgG content were observed when

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217 comparing levels before and after centrifugation, indicating that few insoluble aggregates were
218 formed. However, centrifugation of the HTST sample without GI led to a pronounced loss of IgG.
219 Centrifugation of HTST + GI samples led to no change in IgG compared to the non-centrifuged
220 samples (all $P < 0.05$, except for HTST alone).

221 **Stability of proteins in BC samples**

222 Equal amounts of proteins from each sample were separated by SDS PAGE under non-reducing
223 conditions and resolved into several major proteins which were identified based partly on their
224 migration compared to bovine protein standards and also by mass spectrometry of the excised SDS
225 PAGE protein bands (Fig. 2A). IgG was resolved into two bands and when compared to an IgG
226 protein standard, the mobility of these corresponded to IgG1 and IgG2.[12] Compared to untreated
227 liquid BC (RBC 0h), SD of unpasteurized BC (BC00-γ) had no effect on the amounts of IgG1,
228 however GI (BC00+γ) lowered IgG1 (30%, $P < 0.05$, Fig. 2B). Interestingly, LTLT pasteurization
229 led to no loss of IgG1, however following SD and GI, levels of IgG1 were lowered (38%, $P < 0.05$).
230 HTST pasteurization of BC reduced IgG1 further to 40%. These quantities were further affected by
231 GI (to 63%). The effects of processing on IgG2 largely mirrored those of IgG1, however it appeared
232 to be slightly more labile than IgG1. The stability of LF was similar to IgG1 and IgG2, however LF
233 was more labile than IgG1 under LTLT pasteurization (20% loss). In contrast, LP was very labile
234 during processing with quantities being markedly reduced following LTLT (55% loss) and HTST
235 (80% loss) pasteurization ($P <0.05$). GI had no effects on the amounts of LP following HTST
236 pasteurization. Processing of BSA reflected the changes occurring with IgG1 and IgG2. LTLT
237 pasteurization lowered the levels of β-LgA, compared to that of β-LgB. α-La tended not to be as
238 susceptible to the effects of GI as the other proteins.

239 **Stability of proteins in BC as determined by Western blot analysis**

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240 Western blot analysis of IgG largely mirrored those determined by ELISA, with high amounts of
241 IgG in the absence of pasteurization being reduced in accordance with the severity of processing.
242 Compared to unpasteurized BC, no differences in IgG levels were observed following LTLT, nor
243 after LTLT and GI. A reduction in IgG was observed following HTST pasteurization and after
244 HTST plus GI (Fig. 3A). For LF, GI of unpasteurized BC resulted in a non-significant fall in
245 concentrations. No effects of LTLT pasteurization was observed on LF. A significant fall (60%
246 loss) was observed following HTST+GI ($P < 0.05$, Fig. 3B).

247 **Determination of the stability of selected proteins by proteomic analysis**

248 The abundance of selected proteins during processing was studied using label free-quantitative
249 proteomics, following centrifugation of the samples to remove insoluble aggregates. Compared to
250 unpasteurized BC subjected to SD but no GI, (i.e. BC00), levels of β-Lg (results not shown), LF,
251 LP, cathelicidin-1, polymeric immunoglobulin receptor (pIgR) and Serpin F2 were reduced in
252 accordance with the severity of processing, whereas contents of lactadherin, Serpin A3-1 and Serpin
253 C1 were unchanged (Fig. 4A, B, C, D, E, F, G and H). In all cases, LTLT pasteurization, (BC63)
254 followed by SD did not affect the stability of proteins compared with unpasteurized BC (BC00-γ)
255 subjected to SD.

256 **Amino acid (methionine) oxidation in BC samples**

257 Compared with non-pasteurized SD BC (BC00-GI), differences in total methionine oxidation were
258 observed (Fig. 5). Firstly, GI of non-pasteurized SD BC resulted in increased methionine oxidation.
259 Although LTLT-pasteurized BC (BC63-GI) was not significantly different from BC00-GI, GI of
260 LTLT-treated BC (BC63+GI) resulted in an increased methionine oxidation (all $P < 0.05$). No
261 significant differences in total methionine oxidation were observed between BC00-GI and HTST-
262 treated BC (BC72-GI), nor if it was treated with GI (BC72+GI). For selected individual proteins,
263 differences in methionine oxidation were detected. No differences in methionine oxidation were

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observed for α-La, αS1-CN and κ-CN. However, increased methionine oxidation was detected following GI for αS2-CN, β-LG, LF, β-CN, BSA (Table 2).

Additional bioactive proteins present in BC

Based on proteomic analysis using LC-MS/MS, 128 proteins with high or medium FDR confidence were detected in raw, untreated BC. Information of the identified proteins, including their accession number and FDR confidence and order according to abundance are shown in Table 3. Some of the major functions are divided into the following categories: anti-inflammatory/innate immunity proteins (e.g. pIgR), anti-proteinase proteins (e.g. serpins) and antimicrobial proteins (e.g. LF and cathelicidins). Proteins related to other effects were also detected, including proteins related to apoptosis and LPS-binding effects. Several of these proteins have not been described in BC previously.

Discussion

Bovine colostrum is well known to possess bioactive health properties but its use for sensitive human patient groups, both adults and infants, remain poorly studied and not clearly verified. [30] This uncertainty may partly result from highly variable effects of different pre-processing techniques, used to prepare colostrum products for human use. In this study, we document highly significant denaturation of key bioactive milk proteins after two widely accepted pasteurization methods (LTLT, HTST), with or without GI, to further secure microbial safety of a powdered BC product from a commercial supplier. The SD, LTLT product was associated with least damage to bioactive proteins, but damage was increased by subsequent irradiation (e.g. marked oxidation effects). It remains to be investigated if irradiation at lower doses would induce less damage to proteins while preserving the capacity to kill any remaining bacterial contaminants. It is also possible that the abundant presence of many bioactive proteins in BC, also documented by our

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proteomic studies, is sufficient to exert beneficial effects in sensitive patient groups, despite some reductions in the fresh product. Thus, a SD, LTLT-pasteurized BC product has shown highly gut-protective effects in preterm, chemotherapy-treated or gut-resected piglets. [15, 31-33] In corresponding pilot clinical trials in infants, tendencies to gut health improvements have been documented after supplementation with a SD, LTLT-pasteurized product, γ-irradiated after packaging, but firm evidence is still lacking. [16, 18, 34]

BC proteins consist of micellar caseins (CN) and whey proteins. The whey protein β-Lg contains a free thiol, which becomes exposed during thermal treatments, reacting with κ-CN on the micelle surface. α-La subsequently binds to β-Lg by hydrophobic interaction forming an aggregate. Thus, we observed by proteomics that β-Lg was lowered during processing. β-Lg binds not only to itself, but also to other bioactive proteins, such as IG, LF and LP. [35] During NSI analysis, the pH is adjusted to pH 4.6, resulting in isoelectric precipitation of CNs, therefore any bound whey proteins will be removed in the filtrate. Despite the presence of some soluble aggregates in the filtrate, NSI analysis is indicative of general protein denaturation during thermal processing. The so-called FILTERMAT SD (in which the drying of powder from the drier occurs under highly controlled conditions with a conveyor belt) induced only slight protein denaturation. GI further increased denaturation of the non-pasteurized powder, and this may be the result of protein oxidation which promotes interprotein disulphide formation, protein homodimers, mixed dimers and aggregate formation. [36] LTLT pasteurization is close to the major IgG transition temperature (Tm) of denaturation. [37] The mild denaturation resulting from combined LTLT and SD was augmented by GI. Thus, LTLT treatment may induce protein unfolding, facilitating the effects of ionising radiation. HTST pasteurization markedly increased denaturation, despite the short time of heat treatment. This is explained because of the high protein concentration of colostrum and secondly because the temperature reached is beyond the Tm of both the IgG Fab and Fc domains. [38] In

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contrast, the lack of further denaturation by HTST and GI indicates that a maximal degree of denaturation was reached forming non-native states which undergo coalescence and create high molecular weight aggregates, which are removed by centrifugation.

Using ELISA, total IgG (IgG1 and IgG2) reflected changes in protein denaturation by NSI analysis. Interestingly, SD of non-pasteurized liquid colostrum, with or without GI, induced loss of IgG. IgG levels were not changed following centrifugation, indicating no participation of IgG in aggregate formation in these non-pasteurized samples and that the type of SD used does not promote aggregate formation. Although GI reduces the quantity of IgG, there was no difference following centrifugation, indicating that GI alone does not promote insoluble aggregates. LTLT or HTST pasteurization lowered total IgG compared to the non-pasteurized control. In both cases, GI reduced IgG levels, but the most pronounced reduction resulted from GI of HTST-pasteurized BC and spray-dried BC. However, centrifugation of the LTLT samples, with or without GI, resulted in an increase in antigenic IgG, which was not expected. A reason may be aggregates in the non-centrifuged samples having reduced numbers of antigenic epitopes and sterically hindering native IgG molecules from gaining access to the antibody. In contrast, HTST-treatment lowered IgG due to more aggregation reaching a degree where no further effects of GI could be noted.

The main colostrum proteins were separated by SDS PAGE into IgG1 and IgG2, BSA, caseins, β-LgA and β-LgB and α-La. IgG1 is to 10 times greater than IgG2, when analyzed by ELISA. [12] In this study, IgG1 and IgG2 were determined by SDS PAGE and the magnitude of these two proteins corresponded to previous studies. SDS PAGE indicated a similar denaturation behaviour of both IgG isotypes, but IgG2 exhibited a greater susceptibility of denaturation following LTLT and HTST thermal processing than IgG1. Bovine IgG aggregates during conditions simulating LTLT pasteurization, increasing human Fc receptor binding. [21] The denaturation of

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bovine IgG isotypes has not been described in detail earlier, but the C_H3 domains of human IgG1 and IgG2 Fc, which are important in immune responses, differ in their susceptibility to denaturation, with IgG2 being most sensitive. This thermal sensitivity may be explained by the presence two methionine substitutions in IgG2 which are absent in IgG1. [39]

Denaturation of LF and LP followed the same trend as IgG, however, LF was more susceptible than IgG1 and IgG2 to LTLT pasteurization. LF is present as a ferric iron depleted form (apo), an iron-saturated form (holo) and finally as a partially iron-saturated form. The iron saturation level of bovine milk is 10-20% but the iron saturation level of LF in this study is unknown. It is likely that the majority of LF is in the apo form, but with a smaller proportion as holo LF. Apo LF denatures close to 70 °C, in contrast to the more stable holo form which denatures at 90 °C. [4] Thus, apo-LF denatures close to temperatures applied during LTLT and HTST pasteurization. It is interesting that SD did not affect the levels of LF, although GI led to effects similar to IgG.

LP was more affected by processing than LF. Like LF and IgG, LP becomes irreversibly unfolded at 70 °C close to LTLT and HTST pasteurization temperatures. [40] LP has 15 cysteine residues, 12 of which contribute to intra-molecular disulphide bridge formation and thus there are some free cysteine residues available. [41, 42] During processing, LP may form disulphide linked aggregates with β-Lg and casein micelles. BSA completely mirrored the changes occurring with IgG1. LTLT had no effects on β-Lg B but significant effects were noted for β-Lg A, in line with the earlier onset temperature of denaturation of β-Lg A compared to β-Lg B (62 vs. 65 °C), which will result in a more effective exposure of its reactive thiol group and aggregate formation. [43]

Effects on α-La were less pronounced and no differences of LTLT treatment were observed. Only GI applied to LTLT-treated BC lowered the quantities of this protein. HTST

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357 treatment reduced levels further, but no additional effects of GI were observed. α-La chelates
358 divalent and monovalent cations such as calcium and sodium which confer thermostability. [44] α-La
359 has no free thiol group, unlike caseins and LP, and aggregates with proteins such as β-Lg via non-
360 covalent hydrophobic interaction.

361 Western blot analysis of IgG demonstrated a similar trend as those by total IgG ELISA.
362 Levels of IgG were lowered in particular after GI. These effects correspond with an earlier study in
363 which GI reduced concentrations of IgG in colostrum replacer in a linear manner. [45] Results from
364 Western blot analysis of LF were comparible with data from SDS PAGE analysis.

365 Many bioactive proteins were detected in BC by proteomics, the abundance of
366 selected proteins was affected by the severity of processing in accordance with SDS PAGE and
367 Western blot analysis. LF is a known antimicrobial and immune-stimulating protein and may
368 provide protective effects against inflammation and sepsis in preterm infants. [46-48] Significant
369 quantities of LF were maintained despite LTLT pasteurisation but were affected by HTST
370 pasteurisation and also following GI. LP was also lowered following HTST pasteurisation and GI.

371 A large group of antimicrobial peptides, the cathelicidins were detected in BC. These have
372 potent antimicrobial activity due to their amphipathic nature which disrupts membranes of microbes
373 and protozoa. Some cathelicidins also have anti-cancer properties. [49] Cathelicidins are low at birth,
374 particularly in preterm infants, suggesting that their supplementation through BC feeding may
375 protect against neonatal sepsis. [50] Several immune proteins were detected, including β2-
376 microglobulin (a component of MHC-1), polymeric immunoglobulin receptor (pIgR), and low-
377 affinity immunoglobulin gamma Fc region receptor II. These proteins are important for complement
378 activation, opsonisation and antibody-dependent cytotoxicity. [50] The pIgR ectodomain is secretory
379 component, which is cleaved from pIgR and binds to and inhibits the pro-inflammatory cytokine

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380 IL-8 which is released during intestinal inflammatory diseases such as NEC. [51, 52] pIgR is thus an
381 important protein contributing to the anti-inflammatory effects of BC. pIgR levels were only
382 reduced following HTST pasteurisation and HTST + GI. Lactadherin (MFG-E8), a component of
383 the milk fat globule, is protective against sepsis by induction of the clearance of apoptotic cells and
384 a reduction in pro-inflammatory cytokine secretion through NF-κB. [53-58] Lactadherin quantities
385 were not significantly different from the control during processing and this stability may be related
386 to its glycosylation.

387 A major finding of this study is that several anti-proteinases, such as the Serpin A1, Serpin
388 A3 family members (Serpin A3-1, 2, 3, 5, 7, 8), Serpin F1, F2 and Serpin C1, α-2- macroglobulin,
389 Kininogens-1 and -2, Inter-alpha-trypsin inhibitor, Cystatin C, Factor XIIa inhibitor, and protein
390 AMP were detected which inactivate the many proteinases released during the necrotic phase of
391 NEC. For instance, Serpin A3 (antichymotrypsin), is an extracellular protease inhibitor inhibiting
392 cathepsin G. During intestinal inflammation (including NEC), serine proteases released by
393 neutrophils (elastase, chymotrypsin and cathepsin G, Caspase 3 and Caspase 8) and matrix
394 metalloproteinases potentiate tissue damage causing a vicious cycle of apoptosis, tissue destruction,
395 and serpin A3 family members are strongly protective against these. [59-62] Cathepsin G also cleaves
396 the pro-forms of IL-8 into mature IL-8, consequently inhibition of this enzyme by serpin A3 will
397 dampen inflammation. [63] Serpin A3-1 was not affected by processing and may be related to its
398 reported metastability. [64] The Serpin A3 family has been previously observed in tissue from cattle [65]
399 and in bovine milk together with other protease inhibitors including α-2 macroglobulin, protein
400 AMBP, cystatin, intertrypsin inhibitor and serpins A3-7, Serpin-A1, A4, A5, C1, D1 and G1. [66]
401 Serpin F2 was affected by HTST pasteurization and GI, but maintained during LTLT
402 pasteurization. Serpin F2 plays an an important role against severe gram-negative sepsis by
403 reducing bacterial growth, tissue destruction and by providing anti-inflammatory effects and

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enhancing fibrinolysis. [67] Serpin C1 was not affected by any of the applied processing treatments and is protective against NF-κB signaling following ischemia/reperfusion injury. [68]

Potentially important oxidative changes occurred during the processing of BC proteins. Using protein-bound methionine as a marker, we observed that pasteurization and SD were not solely responsible for induction of oxidative damage, but that GI itself greatly induced BC oxidation. Such effects have been reported in soy protein products and also in α- and β-crystallins. [69, 70] Methionine is a hydrophobic amino acid buried in the protein hydrophobic core, but some residues are exposed near to or on the surface and become more rapidly oxidized. For instance, the only methionine residue in α-La is surface-exposed, whereas in β-Lg, two of the four methionine resides are only partly exposed. [36] α-La was not oxidized on methionine in this study, whereas β-CN and αs_2-CN were, likely because of the random structure of these caseins and also their amounts in BC, where caseins are the major proteins (Table 2). Oxidative damage is a common observation in patients, including preterm infants that have difficulties in regulating blood flow and tissue oxygenation in the days and weeks after preterm birth. Because such infants are partly unable to control the increased levels of reactive oxygen species, the high total antioxidant capacity of colostrum may be protective against damaging free oxygen radicals. [11]

In summary, BC contains many proteins with potentially potent effects against inflammation and maintainance of the control against microbiological, immunological and physiological imbalances which can occur during compromized intestinal conditions, such as those present in the newborn and especially the preterm newborn. Several bioactive proteins are highly sensitive to processing procedures and the greatest effects are observed during HTST pasteurization. Further, high-dose GI treatment affects the stability of bioactive proteins in BC, especially when combined with HTST processing.

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Conclusions

LTLT pasteurization followed by SD is an optimal method to preserve the levels of important BC anti-inflammatory proteins when powdered BC is used as a bioactive diet supplement for sensitive patient groups, such as preterm infants, chemotherapy-treated cancer patients, subjects with enteric inflammation or antibiotic-associated diarrhea, or gut surgical patients.

Abbreviations

AGE	Advanced glycosylation end products
α-La	Alpha-lactalbumin
BC	Bovine colostrum
β-Lg	Beta-lactoglobulin
CN	Casein
DM	Human donor milk
FDR	False discovery rate
GI	Gamma irradiation
HRP	Horse radish peroxidase
HTST	High-temperature/sort-time pasteurization (72 °C, 15 s)
IF	Infant formula
LDS	Lithium dodecyl sulfate
LF	Lactoferrin
LP	Lactoperoxidase
LPS	Lipopolysaccharide
LTLT	Low-temperature long-time pasteurization (63 °C, 30 min)
NEC	Necrotizing enterocolotis
NSI	Nitrogen solubility index
PVDF	Polyvinylidene difluoride
SD	Spray-drying
UHT	Ultra-high-temperature processing
TBST	Tris buffered saline, with Tween
Tm	Denaturation midpoint

Author contributions

D. E. W. C., S.A., T.H.H., D.N.N., C.D.G., R.L. and P. T. S. contributed to the concept and design of the study. D.E.W.C and S.A. conducted the study and performed the data analysis. All authors contributed to the data interpretation and manuscript writing. D.E.W.C had primary responsibility for the final content. All authors accepted the final manuscript.

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Conflict of interest statement

No employees of Biofiber Damino (including co-author Tim Hesselballe Hansen) had any influence on the study design nor the interpretation of results. University of Copenhagen holds a patent on the use of BC for pediatric patients. Prof. Sangild is listed as sole inventor but has declined any share of potential revenue arising from commercial exploitation of such a patent. The remaining authors have no conflicts of interest

Acknowledgements

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Figure legends

Figure 1: (A) Degree of denaturation of bovine colostrum samples determined by NSI analysis. RBC 0h: Raw liquid BC in the bulk tank immediately after thawing at 4 °C; RBC 6h: Raw liquid BC in the bulk tank 6h after thawing at 4 °C; BC00-γ: SD BC, no GI; BC00+γ: SD BC, +GI; BC63-γ: LTLT-pasteurized SD BC, no GI; BC63+γ: LTLT-pasteurized SD BC, + GI ; BC72-γ: HTST-pasteurized SD BC, no GI; BC72+γ: HTST-pasteurized SD BC, +GI. The relative protein content of the samples was normalised to the RBC 0h sample. Values represent means ± SD (n = 3). Labels not sharing a common letter are significantly different ($P < 0.05$). **(B) Degree of denaturation of bovine IgG in bovine colostrum samples**. Total IgG was measured by ELISA. BC00-γ: SD BC, no GI; BC00+γ: SD BC, +GI; BC63-γ: LTLT-pasteurized SD BC, no GI; BC63+γ: LTLT-pasteurized SD BC, + GI; BC72-γ: HTST-pasteurized SD BC, no GI; BC72+γ: HTST-pasteurized SD BC, +GI. C: Solubilized SD samples were centrifuged at 10,000 x g for 15 min to removed insoluble aggregates and the supernatant was used for analysis. The relative IgG content of the samples was normalised to the BC00-γ sample. Values represent means ± SD (n = 3). Labels not sharing a common letter are significantly different ($P < 0.05$).

Figure 2: (A) SDS PAGE of liquid and reconstituted bovine colostrum. BC samples were subjected to centrifugation at 10,000 x g for 15 min at 4 °C for removal of insoluble aggregates. For subsequent densitometric purposes, samples were applied randomly to the gel. RBC 0h: Raw liquid BC immediately after thawing at 4 °C; RBC 6h: Raw liquid BC 6h after thawing at 4 °C; BC00-γ: SD BC, no GI; BC00+γ: SD BC, +GI; BC63-γ: LTLT-pasteurized SD BC, no GI; BC63+γ: LTLT-pasteurized SD BC, + GI ; BC72-γ: HTST-pasteurized SD BC, no GI; BC72+γ: HTST-pasteurized SD BC, +GI. Samples were compared to a molecular weight marker (lanes 1 and 8) and BSA (2 µg, lane 2) was used to normalize for Coomassie Blue staining in subsequent densitometric analysis.

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482 **(B) Quantitative analysis of proteins by densitometry.** Band volumes were measured following
483 SDS PAGE of liquid and reconstituted BC after centrifugation at 10,000 x g for 15 min at 4 °C.
484 RBC 0h: Raw liquid BC immediately after thawing at 4 °C; RBC 6h: Raw liquid BC 6h after
485 thawing at 4 °C; BC00-γ: SD BC, no GI; BC00+γ: SD BC, +GI; BC63-γ: LTLT-pasteurized SD
486 BC, no GI; BC63+γ: LTLT-pasteurized SD BC, + GI ; BC72-γ: HTST-pasteurized SD BC, no GI;
487 BC72+γ: HTST-pasteurized SD BC, +GI. The relative protein content of the samples was
488 normalised to the RBC 0h sample. Values represent means ± SD (n = 3). Labels not sharing a
489 common letter are significantly different ($P < 0.05$).

490 **Figure 3: Quantification of bovine colostrum proteins by Western blot.** BC proteins were
491 analyzed by Western blot analysis for IgG (A) and LF (B). For IgG, 0.1 µg protein was applied per
492 lane, whereas for LF, 5 µg was applied per lane. Proteins were resolved by non-reducing SDS
493 PAGE and electrotransferred onto PVDF membranes. Proteins were detected by Western blotting
494 and detected by enhanced chemiluminescence. BC00-γ: SD BC, no GI; BC00+γ: SD BC, +GI;
495 BC63-γ: LTLT-pasteurized SD BC, no GI; BC63+γ: LTLT-pasteurized SD BC, + GI ; BC72-γ:
496 HTST-pasteurized SD BC, no GI; BC72+γ: HTST- pasteurized SD BC, +GI. C: Solubilized SD
497 samples were centrifuged at 10,000 x g for 15 min to removed insoluble aggregates and the
498 supernatant was used for analysis. Values represent means ± SD (n = 3). *: Mean value was
499 significantly different from that of the BC00-γ control ($P < 0.05$).

500 **Figure 4: Levels of selected proteins determined by proteomic analysis.** The abundance of
501 lactoferrin (A); lactoperoxidase (B); cathelicidin-1 (C); polymeric immunoglobulin receptor (pIgR)
502 (D); lactadherin (E) Serpin A3-1 (F); Serpin F2 (G) and Serpin C1 (H) were compared to the
503 severity of processing. Quantification of total ion abundance (TIC) was determined by summing the
504 intensities of the ions derived from peptides released from each protein. BC00: SD BC, no GI;

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BC63: LTLT-pasteurized SD BC, no GI; BC63+γ: LTLT-pasteurized SD BC, + GI ; BC72: HTST-pasteurized SD BC, no GI; BC72+γ: HTST-pasteurized SD BC, +GI. Values represent means ± SD (n = 3). * Mean value was significantly different from that of the BC00 control ($P < 0.05$), n=3; **: ($P < 0.01$).

Figure 5: Total oxidized methionine of bovine colostrum. The total abundance (TIC) of oxidized methionine located on peptides following trypsin treatment of colostrum samples was analyzed by LC-MS/MS. Quantification of total abundance was determined by summing the intensities of the peptides containing the oxidized methionine for each sample. Significance testing was performed using one-way ANOVA, $P < 0.05$, n=3. BC00-γ: SD BC, no GI; BC00+γ: SD BC, +GI; BC63-γ: LTLT-pasteurized SD BC, no GI; BC63+γ: LTLT-pasteurized SD BC, + GI ; BC72-γ: HTST-pasteurized SD BC, no GI; BC72+γ: HTST-pasteurized SD BC, +GI.

Table 1: Treatment of the bovine colostrum samples. Eight different BC samples were subjected to varying pasteurization procedures, with and without spray drying (SD) and gamma irradiation (GI). RBC 0 h: Raw BC sampled from the bulk tank at 0 h after thawing of collected frozen colostrum; RBC 6 h: Raw BC sampled from the bulk tank at 6 h after thawing of collected frozen colostrum; BC00-γ: Non-pastuerized BC, subsequent SD and no GI; BC00+γ: Non-pastuerized, BC subsequent SD and treated with GI; BC63-γ: BC subjected to LTLT pasteurization (63 °C, 30 min), subsequent SD and no GI; BC63+γ: Bovine colostrum subjected to LTLT pasteurization (63 °C, 30 min), subsequent SD and treated with GI; BC72-γ: BC subjected to HTST pasteurization (72 °C, 15 s), subsequent SD and no GI; BC72+γ: BC subjected to HTST pasteurisation (72 °C, 15 s), subsequent SD and treated with GI.

Table 2: Methionine oxidation of some major bovine colostrum proteins. The total abundance (TIC) of oxidized methionine located on BC proteins following trypsin treatment of reduced and

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alkylated proteins was analyzed by LC-MS/MS. Quantification of total abundance was determined by summing the abundance (TIC) of ions containing the oxidized methionine in each BC sample. * Mean value was significantly different from that of the BC00 control ($P < 0.05$), n=3. BC00: SD BC, no GI; BC00+γ: SD BC, +GI; BC63-γ: LTLT-pasteurized SD BC, no GI; BC63+γ: LTLT-pasteurized SD BC, + GI ; BC72-γ: HTST-pasteurized SD BC, no GI; BC72+γ: HTST-pasteurized SD BC, +GI; N.S.: No significant differences between treatments.

Table 3: Identification of proteins present in raw bovine colostrum. Proteins were identified from the total abundance (TIC) of peptides released by trypsin treatment of reduced and alkylated proteins from raw bovine colostrum.

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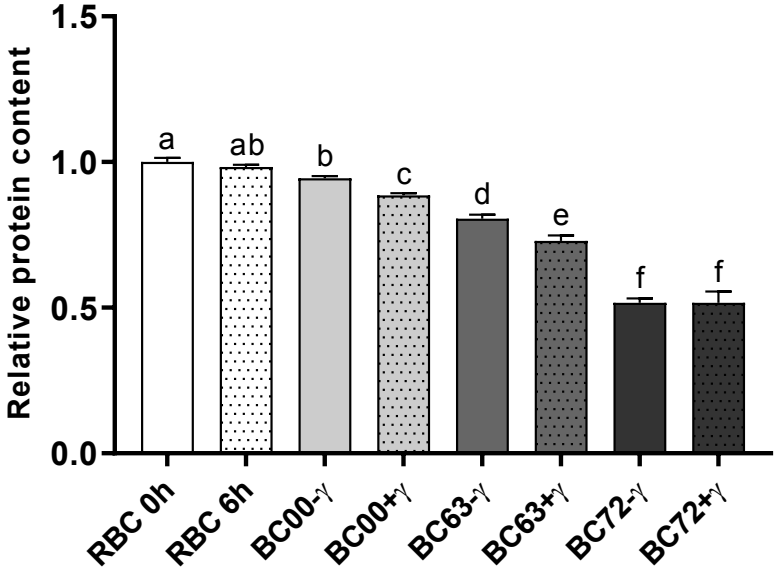
Figure 1

(A)



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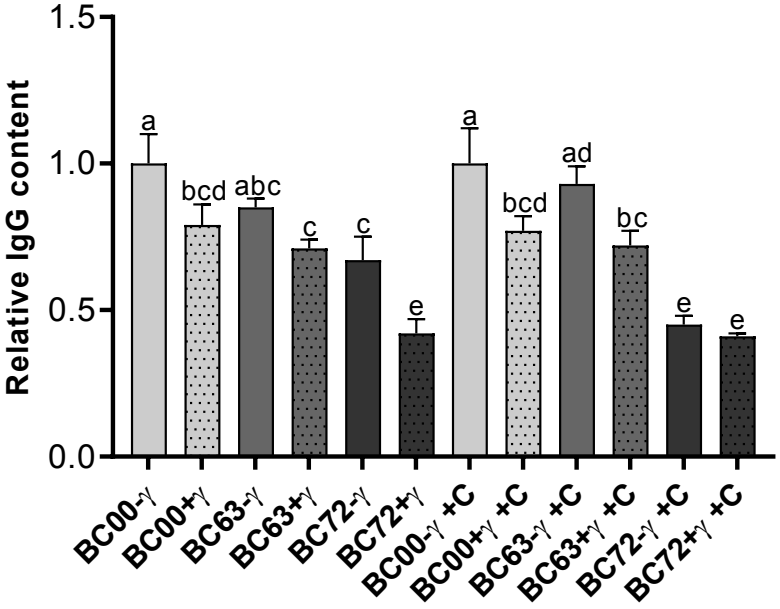

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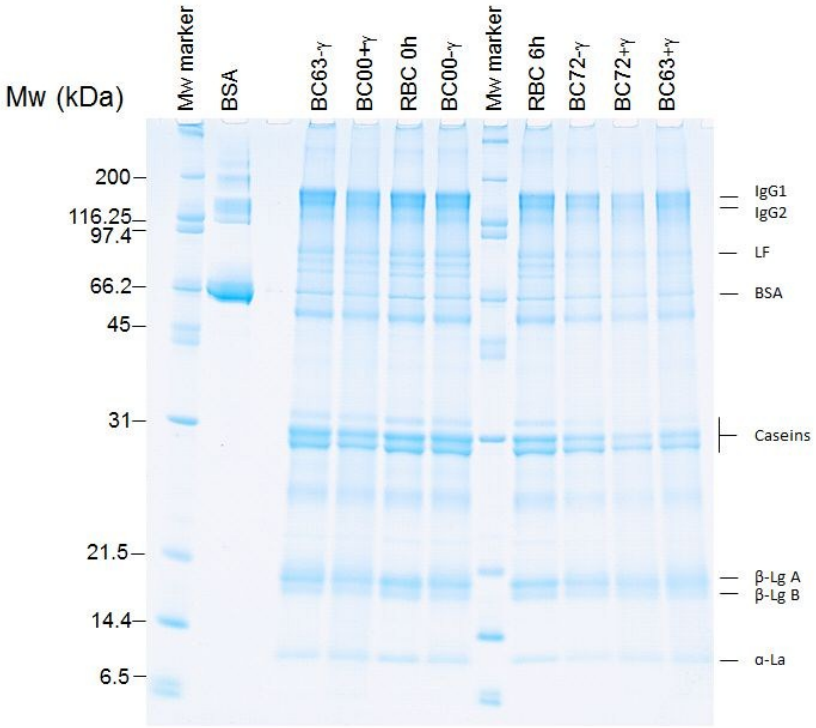
Figure 2

(A)



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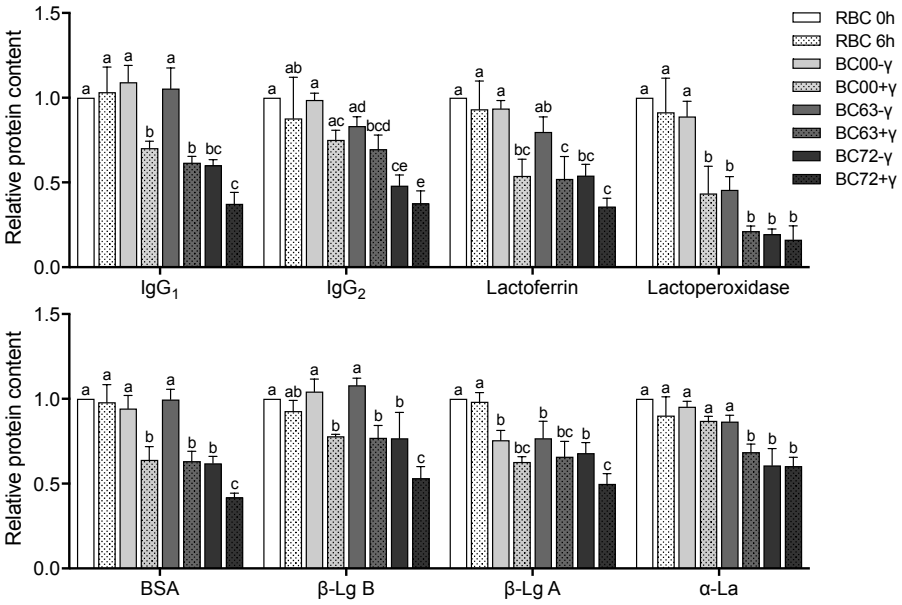

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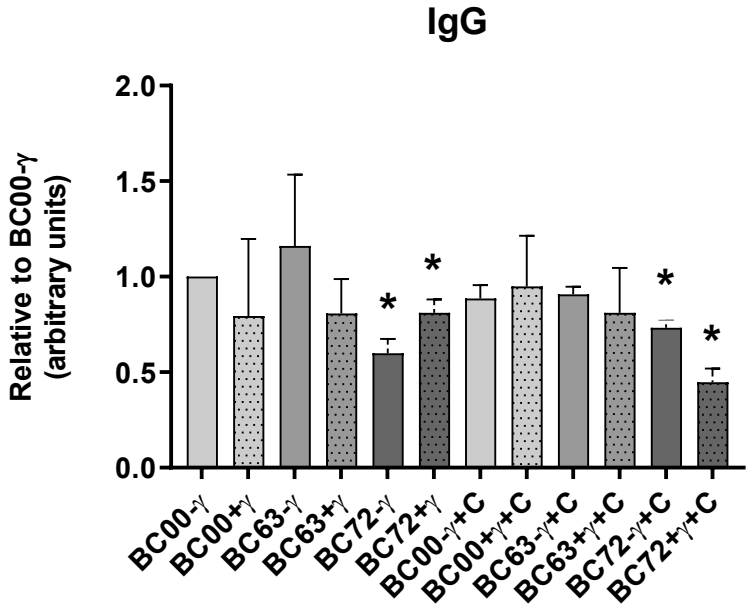
Figure 3

(A)



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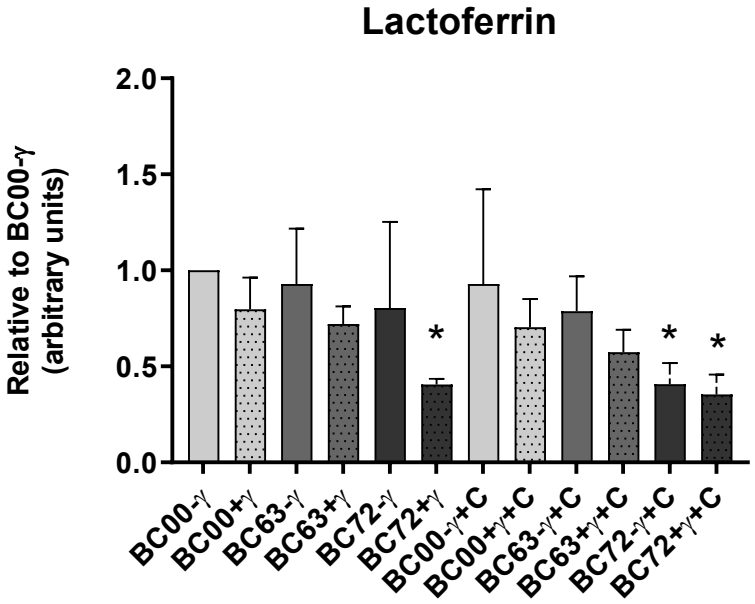

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Figure 4

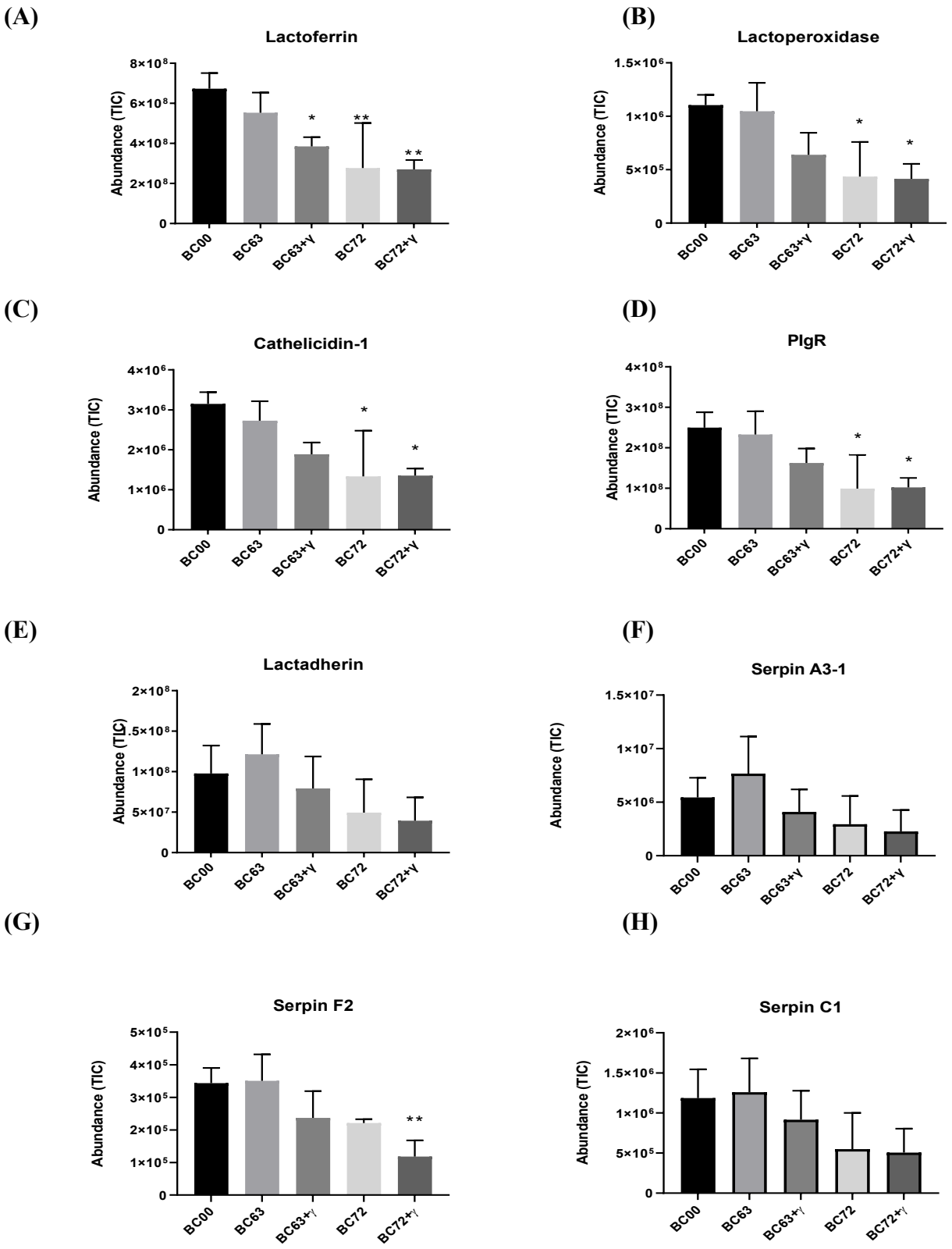

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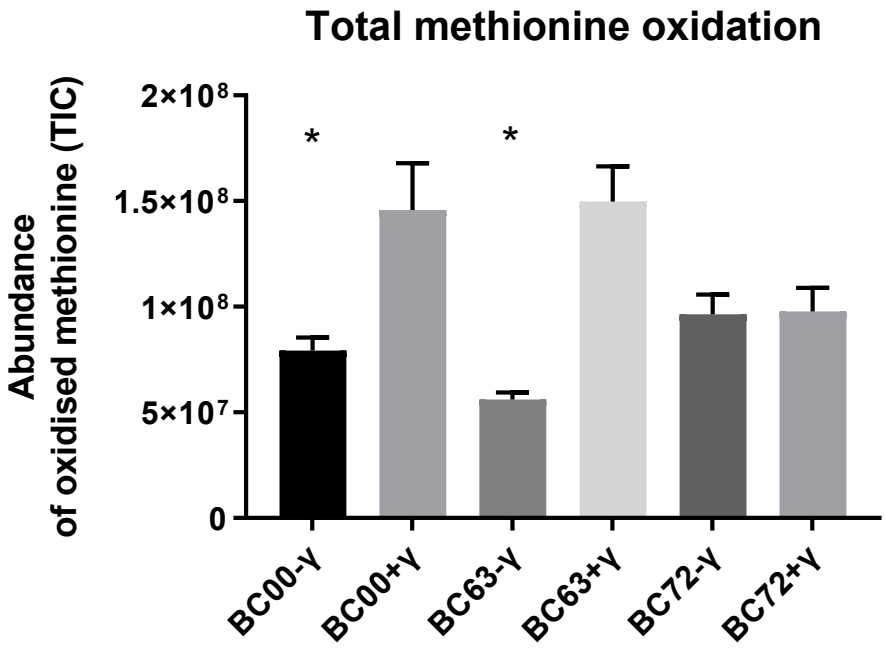
Figure 5



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560 **Table 1**

561

562

563

Sample	Pasteurization	Spray-drying	γ-irradiation
RBC 0h	None	-	-
RBC 6h	None	-	-
BC00-γ	None	Gentle	-
BC00+γ	None	Gentle	+
BC63-γ	63 °C, 30 min.	Gentle	-
BC63+γ	63 °C, 30 min.	Gentle	+
BC72-γ	72 °C, 15 sec.	Conventional	-
BC72+γ	72 °C, 15 sec.	Conventional	+

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Table 2

Methionine oxidation of selected bovine colostrum proteins.

BC00 vs.	β-CN	αs2-CN	K-CN	αs1-CN	α-La	β-LG	LF	BSA
BC00+ γ	*	*	NS	NS	NS	*	*	NS
BC63	NS	NS	NS	NS	NS	NS	NS	*
BC63+ γ	*	*	NS	NS	NS	NS	**	NS
BC72	*	*	NS	NS	NS	NS	NS	NS
BC72+ γ	*	*	NS	NS	NS	NS	NS	*

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588 **Table 3**

589 **Proteins in raw bovine colostrum determined by LC-MS/MS.**

590 **Proteins with a high, or medium protein FDR confidence are included and are ordered according to abundance.**
591 **Abundances represent the mean of three replicates.**
592

Protein FDR Confidence	Uniprot Accession No.	Description	Abundance	Proposed Bioactive Function	Relevant References
High	P02754	Beta-lactoglobulin	5400305895	(a)	71
High	P02668	Kappa-casein	2382142259	(c)	72-76
High	P02662	Alpha-S1-casein	1065638764	(b,c,d)	72, 76-78
High	P02663	Alpha-S2-casein	941494413	(b,c,e)	72, 76, 78, 79
High	P24627	Lactotransferrin	670929284	(f,g)	47, 48, 80-82
High	P02769	Serum albumin	649493154	(a)	71
High	P02666	Beta-casein	524007205	(b,c,h)	72, 76, 78, 83-86
Medium	A4FUD1	GPN-loop GTPase 1	358167313		
High	Q29443	Serotransferrin	284050732		87, 88
High	P81265	Polymeric immunoglobulin receptor	249692756	(c,f,g)	52, 89, 90
High	P01888	Beta-2-microglobulin	161507711	(f,h)	91-93
High	P00711	Alpha-lactalbumin	155165376	(a,c)	71, 94-96
High	Q95114-1	Lactadherin	97630378	(g,i)	58
High	Q2UVX4	Complement C3	92287964	(i)	97, 98
High	Q95114-2	Isoform Short of Lactadherin	79008125	(g,i)	58
High	P81265-2	Isoform Short of Polymeric immunoglobulin receptor	75890616	(c,f,g)	89, 90
High	Q3ZCH5	Zinc-alpha-2-glycoprotein	55471008	(g,j)	99, 100
High	P17697	Clusterin	44345110	(c,f)	101
High	P18892	Butyrophilin subfamily 1, member A1	40335726		
Medium	P25930	C-X-C chemokine receptor type 4	37562345		
High	Q3SZ57	Alpha-fetoprotein	27183051		
High	Q3SZV7	Hemopexin	24279302	(a,c,k)	102-105
High	P08037	Beta-1,4-galactosyltransferase 1	20214000		
High	P12763	Alpha-2-HS-glycoprotein (Fetuin A)	20087619	(k)	106
High	P80457	Xanthine dehydrogenase/oxidase	17507126		
High	Q7SIH1	Alpha-2-macroglobulin	17056365	(k)	
Medium	E1BE02	Transcription factor E2F7	15701037		
High	P02672	Fibrinogen alpha chain	14142730	(l)	107
High	P34955	Alpha-1-antiproteinase (Serpin A1)	13774037	(c,g,i,k)	68, 108
Medium	Q863H2	Natural cytotoxicity triggering receptor 1	12196129		
High	Q3MHN5	Vitamin D-binding protein	11937188		100
High	Q2KJF1	Alpha-1B-glycoprotein	10656431	(c,m)	109
High	P02676	Fibrinogen beta chain	10100958	(l)	107

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594 **Table 3 (Contd.)**

Protein FDR Confidence	Uniprot Accession No.	Description	Abundance	Proposed Bioactive Function	Relevant References
High	A2I7N3	Serpin A3-7	9488213	(c,g,i,k)	108
High	P10152	Angiogenin-1	8179228	(g)	
High	P80195	Glycosylation-dependent cell adhesion molecule 1	7455473	(f)	110-112
High	P61823	Ribonuclease pancreatic	7024961		
High	Q95122	Monocyte differentiation antigen CD14	6627102	(g,n)	113, 114
High	Q8SPP7	Peptidoglycan recognition protein 1	5868230	(c,f,o)	115-119
High	Q9TUM6	Perilipin-2	5804348		
High	P02702	Folate receptor alpha	5697779		
High	A2I7M9	Serpin A3-2	5429528	(c,g,i,k)	108
High	Q9TTE1	Serpin A3-1	5429528	(c,g,i,k)	108
High	P12799	Fibrinogen gamma-B chain	5427955	(l)	
Medium	Q9MZ06	Fibroblast growth factor-binding protein 1	5089492		
High	O46375	Transthyretin	4937483		
High	P02070	Hemoglobin subunit beta	4662626		
Medium	Q3T0R7	3-ketoacyl-CoA thiolase, mitochondrial	4514163		
High	P17690	Beta-2-glycoprotein 1 (Apolipoprotein H)	4272876	(i,p)	120
High	P50448	Factor XIIa inhibitor (Serpin G1 ortholog)	4231989	(g,k)	
High	P81187	Complement factor B	4212745	(f,g)	121, 122
High	P00735	Prothrombin	4162826		107
High	P15467	Ribonuclease 4	4074023	(f,g)	123
High	P26201	Platelet glycoprotein 4	3972730		
High	Q03247	Apolipoprotein E	3966333	(i,p)	124, 125
High	P06868	Plasminogen	3928148		
High	P01966	Hemoglobin subunit alpha	3805067		
High	P15497	Apolipoprotein A-I	3803419	(i,p)	125
High	Q0VCX2	78 kDa glucose-regulated protein	3709680		
High	Q28110	Low affinity immunoglobulin gamma Fc region receptor II	3481040	(g)	126
High	Q3SX14	Gelsolin	3472389	(q)	127, 128
Medium	P21809	Biglycan	3420513		
High	Q3T052	Inter-alpha-trypsin inhibitor heavy chain H4 (Serpin family member)	3322315	(c,g,i,k)	129-131
High	P22226	Cathelicidin-1	3148987	(f,g)	132, 133
High	P62992	Ubiquitin-40S ribosomal protein S27a, polyubiquitin-B & C, Ubiquitin-60S ribosomal protein L40	3034359	(r)	

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597 **Table 3 (Contd.)**

Protein FDR Confidence	Uniprot Accession No.	Description	Abundance	Proposed Bioactive Function	Relevant References
High	P60712	Actin, cytoplasmic 1,2	3030441	(s)	134
High	Q2KIT0	Protein HP-20 homolog	2806215		
High	F1N152	Serine protease HTRA1	2742640		
High	O02853	Prostaglandin-H2 D-isomerase	2433559		
High	A2I7N2	Serpin A3-6	2237878	(c,g,i,k)	108
High	P07589-1	Fibronectin	2221418		
High	Q3MHN2	Complement component C9	2206168	(g)	
High	P01035	Cystatin-C	2173037	(k)	135
High	Q2KIU3	Protein HP-25 homolog 2	1841849		
High	P39873	Brain ribonuclease	1796393		
High	P30922	Chitinase-3-like protein 1	1749494		
High	P02081	Hemoglobin fetal subunit beta	1419616		
High	P79345	Epididymal secretory protein E1	1356066		
High	Q2M2T1	Histone H2B type 1-K; H2B type1	1304439	(f,g)	136-141
High	P01044-2	Kininogen 1; Isoform LMW of Kininogen-1	1263466	(k,l)	107, 142
High	Q3SZR3	Alpha-1-acid glycoprotein	1208486	(c,m)	109
High	P41361	Antithrombin-III (Serpin C1)	1185626	(c,g,i,k)	68, 143
High	P11151	Lipoprotein lipase	1129259		
High	P62803	Histone H4	1113619		136-141
High	Q32L48	Histone H2B type 1-N	1086706		
High	P80025	Lactoperoxidase	1076184	(c,f)	144-146
High	Q95121	Pigment epithelium-derived factor (Serpin F1)	1049595	(c,g,i,k)	147, 148
High	P13753	BOLA class I histocompatibility antigen, alpha chain BL3-7	995477		
High	O18836	Growth/differentiation factor 8	928508		
High	Q2TBI0	Lipopolysaccharide-binding protein	862472	(g,p)	
High	Q2KIX7	Protein HP-25 homolog 1	840676		
High	Q27965	Heat shock 70 kDa protein 1B; protein 2, cognate 71 kDa protein; 70 kDa protein 1A; 70 kDa protein 1A	831000		
High	P01045	Kininogen-2	804084	(k,l)	107, 142
High	P68138	Actin, alpha skeletal muscle	803904	(s)	134
High	P48034	Aldehyde oxidase 1	780635		
High	Q10741	Disintegrin and metalloproteinase domain-containing protein 10 (ADAM 10)	768663		
High	A2I7N1	Serpin A3-5	711441	(c,g,i,k)	108

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600 **Table 3 (Contd.)**

Protein FDR Confidence	Uniprot Accession No.	Description	Abundance	Proposed Bioactive Function	Relevant References
High	Q2KIS7	Tetranectin	690340	(k)	149
High	A6QPQ2	Serpin A3-8	688712	(c,g,i,k)	108
High	Q2TBU0	Haptoglobin	641649	(f)	
High	Q6R8F2	Cadherin-1	565114		
High	Q29437	Primary amine oxidase, liver isozyme	552552		
High	O18738	Dystroglycan	548444		
High	Q58D62	Fetuin-B	516580	(c,g,k)	150
High	P0C0S9	Histone H2A types 1, 2-C, J	511556	(f,g)	136-141
Medium	P84227	Histone H3 (3.2, 3.3C, 3.3 & 3.1)	431582	(f,g)	136-141
Medium	P00978	Protein AMBP (Inter-alpha-trypsin inhibitor related/Mast cell protease inhibitor)	429100	(c,g,i,k)	129-131, 151
High	Q0P569	Nucleobindin-1	413969		
Medium	A3KN33	Pikachurin	391252		
Medium	Q0VCM5	Inter-alpha-trypsin inhibitor heavy chain H1(Serpin family member)	376625	(c,g,i,k)	129-131
Medium	P19660	Cathelicidin-2	346372	(f,g)	152, 153
High	P28800	Alpha-2-antiplasmin (Serpin F2)	343451	(c,g,i,k)	67
High	Q29S21	Keratin, type II cytoskeletal 7,75,5	248473		
High	P01030	Complement C4	220857	(g)	97
Medium	Q9TTK6	Membrane primary amine oxidase	180341		
High	Q3ZBZ1	45 kDa calcium-binding protein	166121		
High	Q32PJ2	Apolipoprotein A-IV	156077	(i,p)	125
Medium	P68103	Elongation factor 1-alpha 1	155659		
High	P81287	Annexin A5	141015		
Medium	P30932	CD9 antigen	87905		
Medium	P58875	SEC14-like protein 2	79387		
Medium	P46193	Annexin A1	54403		

a: Protection against reactive oxygen species; **b:** Calcium uptake; **c:** Anti-inflammatory; **d:** Proliferative; **e:** Barrier effects through tight junction formation; **f:** Antimicrobial; **g:** Innate immunity; **h:** Chemotactic activity; **i:** Anti-sepsis; **j:** RNase activity; **k:** Protection against proteinases; **l:** Potential fibrin entrappment of microbes and subsequent clearance; **m:** Inhibition of neutrophil activation and complement activity; **n:** Innate immune responses as soluble CD14; **o:** Protection against colitis; **p:** Binding of LPS and lipoteichoic acid; **q:** Regulation of actin, cell survival and inflammation; **r:** Cellular regulation of membrane trafficking and protein degradation; **s:** Protection versus degradation of antimicrobial cathelicidins;

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DOI: 10.1039/C9FO02998B

606 **References**

607 1. Chatterton DE, Nguyen DN, Bering SB, Sangild PT. Anti-inflammatory mechanisms
608 of bioactive milk proteins in the intestine of newborns. Int J Biochem Cell Biol. 2013;45(8):1730-47.
609 2. Pollock JR, Goff HM. Lactoperoxidase-catalyzed oxidation of thiocyanate ion: a
610 carbon-13 nuclear magnetic resonance study of the oxidation products. Biochim Biophys Acta.
611 1992;1159(3):279-85.
612 3. Almehdar HA, El-Fakharany EM, Uversky VN, Redwan EM. Disorder in milk proteins:
613 structure, functional disorder, and biocidal potentials of lactoperoxidase. Curr Protein Pept Sci.
614 2015;16(4):352-65.
615 4. Wang B, Timilsena YP, Blanch E, Adhikari B. Lactoferrin: Structure, function,
616 denaturation and digestion. Crit Rev Food Sci Nutr. 2017:1-17.
617 5. Lonnerdal B. Bioactive Proteins in Human Milk-Potential Benefits for Preterm Infants.
618 Clin Perinatol. 2017;44(1):179-91.
619 6. Li Y, Nguyen DN, Obelitz-Ryom K, Andersen AD, Thymann T, Chatterton DEW, et al.
620 Bioactive Whey Protein Concentrate and Lactose Stimulate Gut Function in Formula-fed Preterm
621 Pigs. J Pediatr Gastroenterol Nutr. 2018;66(1):128-34.
622 7. Drapala KP, Mulvihill DM, O'Mahony JA. Improving the oxidative stability of model
623 whey protein hydrolysate-based infant formula emulsions with lecithin. International Journal of
624 Dairy Technology. 2018;71(4):966-74.
625 8. Verma N, Manna SK. Advanced glycation end products (AGE) potentiates cell death
626 in p53 negative cells via upregulaion of NF-kappa B and impairment of autophagy. J Cell Physiol.
627 2017;232(12):3598-610.
628 9. Silva E, Rangel AHN, Murmam L, Bezerra MF, de Oliveira JPF. Bovine colostrum:
629 benefits of its use in human food. Food Science and Technology. 2019;39:355-62.
630 10. Ceniti C, Froiio F, Britti D, Paolino D, Costanz N. Rheological characteristics of
631 bovine colostrum and their correlation with immunoglobulin G. International Journal of Dairy
632 Technology. 2019;72(3):345-9.
633 11. Zhao J, Liu XJ, Ma JW, Zheng RL. DNA damage in healthy term neonate. Early Hum
634 Dev. 2004;77(1-2):89-98.
635 12. Samarutel J, Baumrucker CR, Gross JJ, Dechow CD, Bruckmaier RM. Quarter
636 variation and correlations of colostrum albumin, immunoglobulin G1 and G2 in dairy cows. J Dairy
637 Res. 2016;83(2):209-18.
638 13. Levieux D, Ollier A. Bovine immunoglobulin G, beta-lactoglobulin, alpha-lactalbumin
639 and serum albumin in colostrum and milk during the early post partum period. J Dairy Res.
640 1999;66(3):421-30.
641 14. Moller HK, Thymann T, Fink LN, Frokiaer H, Kvistgaard AS, Sangild PT. Bovine
642 colostrum is superior to enriched formulas in stimulating intestinal function and necrotising
643 enterocolitis resistance in preterm pigs. Br J Nutr. 2011;105(1):44-53.
644 15. Sty AC, Sangild PT, Skovgaard K, Thymann T, Bjerre M, Chatterton DE, et al. Spray
645 Dried, Pasteurised Bovine Colostrum Protects Against Gut Dysfunction and Inflammation in
646 Preterm Pigs. J Pediatr Gastroenterol Nutr. 2016;63(2):280-7.
647 16. Juhl SM, Ye X, Zhou P, Li Y, Iyore EO, Zhang L, et al. Bovine Colostrum for Preterm
648 Infants in the First Days of Life: A Randomized Controlled Pilot Trial. J Pediatr Gastroenterol Nutr.
649 2018;66(3):471-8.
650 17. Ahnfeldt AM, Hyldig N, Li Y, Kappel SS, Aunsholdt L, Sangild PT, et al. FortiColos - a
651 multicentre study using bovine colostrum as a fortifier to human milk in very preterm infants: study
652 protocol for a randomised controlled pilot trial. Trials. 2019;20(1):279.
653 18. Rathe M, De Pietri S, Wehner PS, Frandsen TL, Grell K, Schmiegelow K, et al.
654 Bovine Colostrum Against Chemotherapy-Induced Gastrointestinal Toxicity in Children With Acute

Published on 25 February 2020. Downloaded on 2/26/2020 7:17:20 AM.

Food & Function Accepted Manuscript

View Article Online
DOI: 10.1039/C9FO02998B

Lymphoblastic Leukemia: A Randomized, Double-Blind, Placebo-Controlled Trial. JPEN J Parenter Enteral Nutr. 2019.

19. Escuder-Vieco D, Espinosa-Martos I, Rodriguez JM, Corzo N, Montilla A, Siegfried P, et al. High-Temperature Short-Time Pasteurization System for Donor Milk in a Human Milk Bank Setting. Front Microbiol. 2018;9:926.

20. Escuder-Vieco D, Espinosa-Martos I, Rodriguez JM, Fernandez L, Pallas-Alonso CR. Effect of HTST and Holder Pasteurization on the Concentration of Immunoglobulins, Growth Factors, and Hormones in Donor Human Milk. Front Immunol. 2018;9:2222.

21. Kulczycki A, Jr. Bovine IgG can aggregate at conditions simulating pasteurization and binds to some human Fc gamma receptors. Mol Immunol. 1987;24(3):259-66.

22. Ford JE, Law BA, Marshall VM, Reiter B. Influence of the heat treatment of human milk on some of its protective constituents. J Pediatr. 1977;90(1):29-35.

23. Evans TJ, Ryley HC, Neale LM, Dodge JA, Lewarne VM. Effect of storage and heat on antimicrobial proteins in human milk. Arch Dis Child. 1978;53(3):239-41.

24. Hong YH, Park JY, Park JH, Chung MS, Kwon KS, Chung K, et al. Inactivation of Enterobacter sakazakii, Bacillus cereus, and Salmonella typhimurium in powdered weaning food by electron-beam irradiation. Radiation Physics and Chemistry. 2008;77(9):1097-100.

25. Osaili TM, Al-Nabulsi AA, Shaker RR, Ayyash MM, Olaimat AN, Abu Al-Hasan AS, et al. Effects of extended dry storage of powdered infant milk formula on susceptibility of Enterobacter sakazakii to hot water and ionizing radiation. Journal of Food Protection. 2008;71(5):934-9.

26. Portela JB, Coimbra PT, Cappato LP, Alvarenga VO, Oliveira RBA, Pereira KS, et al. Predictive model for inactivation of salmonella in infant formula during microwave heating processing. Food Control. 2019;104:308-12.

27. Organization WH. Safety and nutritional adequacy of irradiated food. In: Organization WH, editor. Geneva1994.

28. Chatterton DEW, Rasmussen JT, Heegaard CW, Sorensen ES, Petersen TE. In vitro digestion of novel milk protein ingredients for use in infant formulas: Research on biological functions. Trends in Food Science & Technology. 2004;15(7-8):373-83.

29. De Gobba C, Olsen K, Skibsted LH. Components of wheat flour as activator of commercial enzymes for bread improvement. European Food Research and Technology. 2016;242(10):1647-54.

30. Rathe M, Muller K, Sangild PT, Husby S. Clinical applications of bovine colostrum therapy: a systematic review. Nutr Rev. 2014;72(4):237-54.

31. Shen RL, Thymann T, Ostergaard MV, Stoy AC, Krych L, Nielsen DS, et al. Early gradual feeding with bovine colostrum improves gut function and NEC resistance relative to infant formula in preterm pigs. Am J Physiol Gastrointest Liver Physiol. 2015;309(5):G310-23.

32. Sangild PT, Shen RL, Pontoppidan P, Rathe M. Animal models of chemotherapy-induced mucositis: translational relevance and challenges. Am J Physiol Gastrointest Liver Physiol. 2018;314(2):G231-G46.

33. Nagy ES, Paris MC, Taylor RG, Fuller PJ, Sourial M, Justice F, et al. Colostrum protein concentrate enhances intestinal adaptation after massive small bowel resection in juvenile pigs. J Pediatr Gastroenterol Nutr. 2004;39(5):487-92.

34. Aunsholt L, Jeppesen PB, Lund P, Sangild PT, Ifaoui IB, Qvist N, et al. Bovine colostrum to children with short bowel syndrome: a randomized, double-blind, crossover pilot study. JPEN J Parenter Enteral Nutr. 2014;38(1):99-106.

35. Nguyen DN, Sangild PT, Li Y, Bering SB, Chatterton DEW. Processing of whey modulates proliferative and immune functions in intestinal epithelial cells. J Dairy Sci. 2016;99(2):959-69.

36. Kramer AC, Torreggiani A, Davies MJ. Effect of Oxidation and Protein Unfolding on Cross-Linking of beta-Lactoglobulin and alpha-Lactalbumin. J Agric Food Chem. 2017;65(47):10258-69.

Published on 25 February 2020. Downloaded on 2/26/2020 7:17:20 AM.

Food & Function Accepted Manuscript

View Article Online
DOI: 10.1039/C9FO02998B

37. Vermeer AW, Norde W. The thermal stability of immunoglobulin: unfolding and aggregation of a multi-domain protein. Biophys J. 2000;78(1):394-404.
38. Ionescu RM, Vlasak J, Price C, Kirchmeier M. Contribution of variable domains to the stability of humanized IgG1 monoclonal antibodies. J Pharm Sci. 2008;97(4):1414-26.
39. Latypov RF, Hogan S, Lau H, Gadgil H, Liu D. Elucidation of acid-induced unfolding and aggregation of human immunoglobulin IgG1 and IgG2 Fc. J Biol Chem. 2012;287(2):1381-96.
40. Boscolo B, Leal SS, Ghibaudi EM, Gomes CM. Lactoperoxidase folding and catalysis relies on the stabilization of the alpha-helix rich core domain: a thermal unfolding study. Biochim Biophys Acta. 2007;1774(9):1164-72.
41. DeGioia L, Ghibaudi EM, Laurenti E, Salmona M, Ferrari RP. A theoretical three-dimensional model for lactoperoxidase and eosinophil peroxidase, built on the scaffold of the myeloperoxidase X-ray structure. Journal of Biological Inorganic Chemistry. 1996;1(5):476-85.
42. Wolf SM, Ferrari RP, Traversa S, Biemann K. Determination of the carbohydrate composition and the disulfide bond linkages of bovine lactoperoxidase by mass spectrometry. Journal of Mass Spectrometry. 2000;35(2):210-7.
43. Panick G, Malessa R, Winter R. Differences between the pressure- and temperature-induced denaturation and aggregation of beta-lactoglobulin A, B, and AB monitored by FT-IR spectroscopy and small-angle X-ray scattering. Biochemistry. 1999;38(20):6512-9.
44. Permyakov EA, Permyakov SE, Breydo L, Redwan EM, Almehdar HA, Uversky VN. Disorder in Milk Proteins: alpha-Lactalbumin. Part C. Peculiarities of Metal Binding. Curr Protein Pept Sci. 2016;17(8):735-45.
45. Campbell JM, Russell LE, Crenshaw JD, Weaver EM, Godden S, Quigley JD, et al. Impact of irradiation and immunoglobulin G concentration on absorption of protein and immunoglobulin G in calves fed colostrum replacer. J Dairy Sci. 2007;90(12):5726-31.
46. Kruzel ML, Zimecki M, Actor JK. Lactoferrin in a Context of Inflammation-Induced Pathology. Front Immunol. 2017;8:1438.
47. Pammi M, Suresh G. Enteral lactoferrin supplementation for prevention of sepsis and necrotizing enterocolitis in preterm infants. Cochrane Database Syst Rev. 2017;6:CD007137.
48. Manzoni P, Rinaldi M, Cattani S, Pugni L, Romeo MG, Messner H, et al. Bovine lactoferrin supplementation for prevention of late-onset sepsis in very low-birth-weight neonates: a randomized trial. JAMA. 2009;302(13):1421-8.
49. Kosciuczuk EM, Lisowski P, Jarczak J, Strzalkowska N, Jozwik A, Horbanczuk J, et al. Cathelicidins: family of antimicrobial peptides. A review. Mol Biol Rep. 2012;39(12):10957-70.
50. Schuller SS, Kramer BW, Villamor E, Spittler A, Berger A, Levy O. Immunomodulation to Prevent or Treat Neonatal Sepsis: Past, Present, and Future. Front Pediatr. 2018;6:199.
51. Lodha A, Howlett A, Ahmed T, Moore AM. The Role of Interleukin-6 and Interleukin-8 Circulating Cytokines in Differentiating between Feeding Intolerance and Necrotizing Enterocolitis in Preterm Infants. Am J Perinatol. 2017;34(13):1286-92.
52. Marshall LJ, Perks B, Ferkol T, Shute JK. IL-8 released constitutively by primary bronchial epithelial cells in culture forms an inactive complex with secretory component. J Immunol. 2001;167(5):2816-23.
53. Hansen LW, Khader A, Yang WL, Jacob A, Chen T, Nicastro JM, et al. Deficiency in milk fat globule-epidermal growth factor-factor 8 exacerbates organ injury and mortality in neonatal sepsis. J Pediatr Surg. 2017;52(9):1520-7.
54. Hansen LW, Yang WL, Bolognese AC, Jacob A, Chen T, Prince JM, et al. Treatment with milk fat globule epidermal growth factor-factor 8 (MFG-E8) reduces inflammation and lung injury in neonatal sepsis. Surgery. 2017;162(2):349-57.
55. Yang WL, Sharma A, Zhang F, Matsuo S, Wang Z, Wang H, et al. Milk fat globule epidermal growth factor-factor 8-derived peptide attenuates organ injury and improves survival in sepsis. Crit Care. 2015;19:375.

Published on 25 February 2020. Downloaded on 2/26/2020 7:17:20 AM.

Food & Function Accepted Manuscript

View Article Online
DOI: 10.1039/C9FO02998B

757 56. Shah KG, Wu R, Jacob A, Molmenti EP, Nicastro J, Coppa GF, et al. Recombinant
758 human milk fat globule-EGF factor 8 produces dose-dependent benefits in sepsis. Intensive Care
759 Med. 2012;38(1):128-36.
760 57. Matsuda A, Jacob A, Wu R, Zhou M, Nicastro JM, Coppa GF, et al. Milk fat globule-
761 EGF factor VIII in sepsis and ischemia-reperfusion injury. Mol Med. 2011;17(1-2):126-33.
762 58. Bu HF, Zuo XL, Wang X, Ensslin MA, Koti V, Hsueh W, et al. Milk fat globule-EGF
763 factor 8/lactadherin plays a crucial role in maintenance and repair of murine intestinal epithelium. J
764 Clin Invest. 2007;117(12):3673-83.
765 59. Law RH, Zhang Q, McGowan S, Buckle AM, Silverman GA, Wong W, et al. An
766 overview of the serpin superfamily. Genome Biol. 2006;7(5):216.
767 60. Claud EC. Neonatal Necrotizing Enterocolitis -Inflammation and Intestinal Immaturity.
768 Antiinflamm Antiallergy Agents Med Chem. 2009;8(3):248-59.
769 61. Gagaoua M, Hafid K, Boudida Y, Becila S, Ouali A, Picard B, et al. Caspases and
770 Thrombin Activity Regulation by Specific Serpin Inhibitors in Bovine Skeletal Muscle. Appl Biochem
771 Biotechnol. 2015;177(2):279-303.
772 62. Herrera-Mendez CH, Becila S, Blanchet X, Pelissier P, Delourme D, Coulis G, et al.
773 Inhibition of human initiator caspase 8 and effector caspase 3 by cross-class inhibitory
774 bovSERPINA3-1 and A3-3. FEBS Lett. 2009;583(17):2743-8.
775 63. Padrines M, Wolf M, Walz A, Baggiolini M. Interleukin-8 processing by neutrophil
776 elastase, cathepsin G and proteinase-3. FEBS Lett. 1994;352(2):231-5.
777 64. Lee C, Park SH, Lee MY, Yu MH. Regulation of protein function by native
778 metastability. Proc Natl Acad Sci U S A. 2000;97(14):7727-31.
779 65. Pere-Brissaud A, Blanchet X, Delourme D, Pelissier P, Forestier L, Delavaud A, et al.
780 Expression of SERPINA3s in cattle: focus on bovSERPINA3-7 reveals specific involvement in
781 skeletal muscle. Open Biology. 2015;5(9).
782 66. D'Alessandro A, Zolla L, Scaloni A. The bovine milk proteome: cherishing, nourishing
783 and fostering molecular complexity. An interactomics and functional overview. Mol Biosyst.
784 2011;7(3):579-97.
785 67. Kager LM, Weehuizen TA, Wiersinga WJ, Roelofs JJ, Meijers JC, Dondorp AM, et al.
786 Endogenous alpha2-antiplasmin is protective during severe gram-negative sepsis (melioidosis).
787 Am J Respir Crit Care Med. 2013;188(8):967-75.
788 68. Gatto M, Iaccarino L, Ghirardello A, Bassi N, Pontisso P, Punzi L, et al. Serpins,
789 immunity and autoimmunity: old molecules, new functions. Clin Rev Allergy Immunol.
790 2013;45(2):267-80.
791 69. Horvatic M, Gruner M. [The effect of gamma radiation on the methionine and
792 tryptophan content of soy protein products]. Nahrung. 1993;37(2):147-52.
793 70. Kim I, Saito T, Fujii N, Kanamoto T, Fujii N. One-shot LC-MS/MS analysis of post-
794 translational modifications including oxidation and deamidation of rat lens alpha- and beta-
795 crystallins induced by gamma-irradiation. Amino Acids. 2016;48(12):2855-66.
796 71. Giblin L, Suha Yalcin A, Bicim G, Kramer AC, Chen Z, Callanan MJ, et al. Whey
797 proteins: targets of oxidation, or mediators of redox protection. Free Radic Res. 2019:1-17.
798 72. Trivedi MS, Shah JS, Al-Mughairy S, Hodgson NW, Simms B, Trooskens GA, et al.
799 Food-derived opioid peptides inhibit cysteine uptake with redox and epigenetic consequences. J
800 Nutr Biochem. 2014;25(10):1011-8.
801 73. Huang LG, Zhou W, Rong X, Tao L, Lu WN. [Effects of glycomacropeptide on
802 damage to intestinal tissue and apoptosis of intestinal epithelial cells in neonatal rats with
803 necrotizing enterocolitis]. Zhonghua Er Ke Za Zhi. 2012;50(7):536-42.
804 74. Gustavo Hermes R, Molist F, Francisco Perez J, Gomez de Segura A, Ywazaki M,
805 Davin R, et al. Casein glycomacropeptide in the diet may reduce Escherichia coli attachment to the
806 intestinal mucosa and increase the intestinal lactobacilli of early weaned piglets after an
807 enterotoxigenic E. coli K88 challenge. Br J Nutr. 2013;109(6):1001-12.

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DOI: 10.1039/C9FO02998B

75. Lopez-Posadas R, Requena P, Gonzalez R, Suarez MD, Zarzuelo A, Sanchez de Medina F, et al. Bovine glycomacropeptide has intestinal antiinflammatory effects in rats with dextran sulfate-induced colitis. J Nutr. 2010;140(11):2014-9.
76. Mukhopadhya A, Noronha N, Bahar B, Ryan MT, Murray BA, Kelly PM, et al. Anti-inflammatory effects of a casein hydrolysate and its peptide-enriched fractions on TNFalpha-challenged Caco-2 cells and LPS-challenged porcine colonic explants. Food Sci Nutr. 2014;2(6):712-23.
77. Cairangzhuoma, Yamamoto M, Xijier, Inagaki M, Uchida K, Yamashita K, et al. A preparation of cow's late colostrum fraction containing alphas1-casein promoted the proliferation of cultured rat intestinal IEC-6 epithelial cells. Biosci Biotechnol Biochem. 2013;77(5):992-7.
78. Cosentino S, Gravaghi C, Donetti E, Donida BM, Lombardi G, Bedoni M, et al. Caseinphosphopeptide-induced calcium uptake in human intestinal cell lines HT-29 and Caco2 is correlated to cellular differentiation. J Nutr Biochem. 2010;21(3):247-54.
79. Yasumatsu H, Tanabe S. The casein peptide Asn-Pro-Trp-Asp-Gln enforces the intestinal tight junction partly by increasing occludin expression in Caco-2 cells. Br J Nutr. 2010;104(7):951-6.
80. Nguyen DN, Li Y, Sangild PT, Bering SB, Chatterton DE. Effects of bovine lactoferrin on the immature porcine intestine. Br J Nutr. 2014;111(2):321-31.
81. Reznikov EA, Comstock SS, Hoeflinger JL, Wang M, Miller MJ, Donovan SM. Dietary Bovine Lactoferrin Reduces Staphylococcus aureus in the Tissues and Modulates the Immune Response in Piglets Systemically Infected with S. aureus. Curr Dev Nutr. 2018;2(4):nzy001.
82. Meyer MP, Alexander T. Reduction in necrotizing enterocolitis and improved outcomes in preterm infants following routine supplementation with Lactobacillus GG in combination with bovine lactoferrin. J Neonatal Perinatal Med. 2017;10(3):249-55.
83. Bessette C, Benoit B, Sekkal S, Bruno J, Estienne M, Leonil J, et al. Protective effects of beta-casofensin, a bioactive peptide from bovine beta-casein, against indomethacin-induced intestinal lesions in rats. Mol Nutr Food Res. 2016;60(4):823-33.
84. Plaisancie P, Boutrou R, Estienne M, Henry G, Jardin J, Paquet A, et al. beta-Casein(94-123)-derived peptides differently modulate production of mucins in intestinal goblet cells. J Dairy Res. 2015;82(1):36-46.
85. Barnett MP, McNabb WC, Roy NC, Woodford KB, Clarke AJ. Dietary A1 beta-casein affects gastrointestinal transit time, dipeptidyl peptidase-4 activity, and inflammatory status relative to A2 beta-casein in Wistar rats. Int J Food Sci Nutr. 2014;65(6):720-7.
86. Ul Haq MR, Kapila R, Sharma R, Saliganti V, Kapila S. Comparative evaluation of cow beta-casein variants (A1/A2) consumption on Th2-mediated inflammatory response in mouse gut. Eur J Nutr. 2014;53(4):1039-49.
87. Pu Y, Li S, Xiong H, Zhang X, Wang Y, Du H. Iron Promotes Intestinal Development in Neonatal Piglets. Nutrients. 2018;10(6).
88. Chen AC, Donovan A, Ned-Sykes R, Andrews NC. Noncanonical role of transferrin receptor 1 is essential for intestinal homeostasis. Proc Natl Acad Sci U S A. 2015;112(37):11714-9.
89. Dallas SD, Rolfe RD. Binding of Clostridium difficile toxin A to human milk secretory component. J Med Microbiol. 1998;47(10):879-88.
90. de Oliveira IR, de Araujo AN, Bao SN, Giugliano LG. Binding of lactoferrin and free secretory component to enterotoxigenic Escherichia coli. FEMS Microbiol Lett. 2001;203(1):29-33.
91. Chiou SJ, Wang CC, Tseng YS, Lee YJ, Chen SC, Chou CH, et al. A novel role for beta2-microglobulin: a precursor of antibacterial chemokine in respiratory epithelial cells. Sci Rep. 2016;6:31035.
92. Groves ML, Greenberg R. Bovine homologue of beta2-microglobulin isolated from milk. Biochem Biophys Res Commun. 1977;77(1):320-7.
93. Cejka J, Van Nieuwkoo JA, Mood DW, Kithier K, Radl J. Beta2-microglobulin in human colostrum and milk: effect of breast feeding and physico-chemical characterization. Clin Chim Acta. 1976;67(1):71-8.

Published on 25 February 2020. Downloaded on 2/26/2020 7:17:20 AM.

Food & Function Accepted Manuscript

View Article Online
DOI: 10.1039/C9FO02998B

94. Yamaguchi M, Takai S, Hosono A, Seki T. Bovine milk-derived alpha-lactalbumin inhibits colon inflammation and carcinogenesis in azoxymethane and dextran sodium sulfate-treated mice. Biosci Biotechnol Biochem. 2014;78(4):672-9.
95. Izumi H, Ishizuka S, Inafune A, Hira T, Ozawa K, Shimizu T, et al. alpha-Lactalbumin hydrolysate stimulates glucagon-like peptide-2 secretion and small intestinal growth in suckling rats. J Nutr. 2009;139(7):1322-7.
96. Yamaguchi M, Yoshida K, Uchida M. Novel functions of bovine milk-derived alpha-lactalbumin: anti-nociceptive and anti-inflammatory activity caused by inhibiting cyclooxygenase-2 and phospholipase A2. Biol Pharm Bull. 2009;32(3):366-71.
97. Ricklin D, Hajishengallis G, Yang K, Lambris JD. Complement: a key system for immune surveillance and homeostasis. Nat Immunol. 2010;11(9):785-97.
98. Ricklin D, Reis ES, Mastellos DC, Gros P, Lambris JD. Complement component C3 - The "Swiss Army Knife" of innate immunity and host defense. Immunol Rev. 2016;274(1):33-58.
99. Hassan MI, Waheed A, Yadav S, Singh TP, Ahmad F. Zinc alpha 2-glycoprotein: a multidisciplinary protein. Mol Cancer Res. 2008;6(6):892-906.
100. Senda A, Fukuda K, Ishii T, Urashima T. Changes in the bovine whey proteome during the early lactation period. Anim Sci J. 2011;82(5):698-706.
101. Cunin P, Beauvillain C, Miot C, Augusto JF, Preisser L, Blanchard S, et al. Clusterin facilitates apoptotic cell clearance and prevents apoptotic cell-induced autoimmune responses. Cell Death Dis. 2016;7:e2215.
102. Liang X, Lin T, Sun G, Beasley-Topliffe L, Cavaillon JM, Warren HS. Hemopexin down-regulates LPS-induced proinflammatory cytokines from macrophages. J Leukoc Biol. 2009;86(2):229-35.
103. Lin T, Sammy F, Yang H, Thundivalappil S, Hellman J, Tracey KJ, et al. Identification of hemopexin as an anti-inflammatory factor that inhibits synergy of hemoglobin with HMGB1 in sterile and infectious inflammation. J Immunol. 2012;189(4):2017-22.
104. Rolla S, Ingoglia G, Bardina V, Silengo L, Altruda F, Novelli F, et al. Acute-phase protein hemopexin is a negative regulator of Th17 response and experimental autoimmune encephalomyelitis development. J Immunol. 2013;191(11):5451-9.
105. Wang X, Rojas-Quintero J, Wilder J, Tesfaigzi Y, Zhang D, Owen CA. Tissue Inhibitor of Metalloproteinase-1 Promotes Polymorphonuclear Neutrophil (PMN) Pericellular Proteolysis by Anchoring Matrix Metalloproteinase-8 and -9 to PMN Surfaces. J Immunol. 2019;202(11):3267-81.
106. Hedrich J, Lottaz D, Meyer K, Yiallouros I, Jahnen-Dechent W, Stocker W, et al. Fetuin-A and cystatin C are endogenous inhibitors of human meprin metalloproteases. Biochemistry. 2010;49(39):8599-607.
107. Scharfstein J, Ramos PIP, Barral-Netto M. G Protein-Coupled Kinin Receptors and Immunity Against Pathogens. Adv Immunol. 2017;136:29-84.
108. Kittas C, Aroni K, Matani A, Papadimitriou CS. Immunocytochemical demonstration of a1-Antitrypsin and a1-Antichymotrypsin in human gastrointestinal tract. Hepatogastroenterology. 1982;29(6):275-7.
109. Gemelli C, Martello A, Montanari M, Zanocco Marani T, Salsi V, Zappavigna V, et al. The Orosomucoid 1 protein is involved in the vitamin D - mediated macrophage de-activation process. Exp Cell Res. 2013;319(20):3201-13.
110. Inagaki M, Nakaya S, Nohara D, Yabe T, Kanamaru Y, Suzuki T. The multiplicity of N-glycan structures of bovine milk 18 kda lactophorin (milk GlyCAM-1). Biosci Biotechnol Biochem. 2010;74(2):447-50.
111. Pedersen LR, Hansted JG, Nielsen SB, Petersen TE, Sorensen US, Otzen D, et al. Proteolytic activation of proteose peptone component 3 by release of a C-terminal peptide with antibacterial properties. J Dairy Sci. 2012;95(6):2819-29.
112. Campagna S, Mathot AG, Fleury Y, Girardet JM, Gaillard JL. Antibacterial activity of lactophoricin, a synthetic 23-residues peptide derived from the sequence of bovine milk component-3 of proteose peptone. J Dairy Sci. 2004;87(6):1621-6.

Published on 25 February 2020. Downloaded on 2/26/2020 7:17:20 AM.

Food & Function Accepted Manuscript

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DOI: 10.1039/C9FO02998B

113. Vidal K, Labeta MO, Schiffrin EJ, Donnet-Hughes A. Soluble CD14 in human breast milk and its role in innate immune responses. Acta Odontol Scand. 2001;59(5):330-4.
114. Filipp D, Alizadeh-Khiavi K, Richardson C, Palma A, Paredes N, Takeuchi O, et al. Soluble CD14 enriched in colostrum and milk induces B cell growth and differentiation. Proc Natl Acad Sci U S A. 2001;98(2):603-8.
115. Liu C, Gelius E, Liu G, Steiner H, Dziarski R. Mammalian peptidoglycan recognition protein binds peptidoglycan with high affinity, is expressed in neutrophils, and inhibits bacterial growth. J Biol Chem. 2000;275(32):24490-9.
116. Sharma P, Dube D, Singh A, Mishra B, Singh N, Sinha M, et al. Structural basis of recognition of pathogen-associated molecular patterns and inhibition of proinflammatory cytokines by camel peptidoglycan recognition protein. J Biol Chem. 2011;286(18):16208-17.
117. Saha S, Jing X, Park SY, Wang S, Li X, Gupta D, et al. Peptidoglycan recognition proteins protect mice from experimental colitis by promoting normal gut flora and preventing induction of interferon-gamma. Cell Host Microbe. 2010;8(2):147-62.
118. Royet J, Dziarski R. Peptidoglycan recognition proteins: pleiotropic sensors and effectors of antimicrobial defences. Nat Rev Microbiol. 2007;5(4):264-77.
119. Lu X, Wang M, Qi J, Wang H, Li X, Gupta D, et al. Peptidoglycan recognition proteins are a new class of human bactericidal proteins. J Biol Chem. 2006;281(9):5895-907.
120. Shi T, Iverson GM, Qi JC, Cockerill KA, Linnik MD, Konecny P, et al. Beta 2-Glycoprotein I binds factor XI and inhibits its activation by thrombin and factor XIIa: loss of inhibition by clipped beta 2-glycoprotein I. Proc Natl Acad Sci U S A. 2004;101(11):3939-44.
121. Ballow M, Fang F, Good RA, Day NK. Developmental aspects of complement components in the newborn. The presence of complement components and C3 proactivator (properdin factor B) in human colostrum. Clin Exp Immunol. 1974;18(2):257-66.
122. Nakajima S, Baba AS, Tamura N. Complement system in human colostrum: presence of nine complement components and factors of alternative pathway in human colostrum. Int Arch Allergy Appl Immunol. 1977;54(5):428-33.
123. Murtha MJ, Eichler T, Bender K, Metheny J, Li B, Schwaderer AL, et al. Insulin receptor signaling regulates renal collecting duct and intercalated cell antibacterial defenses. J Clin Invest. 2018;128(12):5634-46.
124. Zhu Y, Kodvawala A, Hui DY. Apolipoprotein E inhibits toll-like receptor (TLR)-3- and TLR-4-mediated macrophage activation through distinct mechanisms. Biochem J. 2010;428(1):47-54.
125. Berbee JF, Havekes LM, Rensen PC. Apolipoproteins modulate the inflammatory response to lipopolysaccharide. J Endotoxin Res. 2005;11(2):97-103.
126. Ulfman LH, Leusen JHW, Savelkoul HFJ, Warner JO, van Neerven RJJ. Effects of Bovine Immunoglobulins on Immune Function, Allergy, and Infection. Front Nutr. 2018;5:52.
127. Roy S, Esmaeilniakooshkghazi A, Patnaik S, Wang Y, George SP, Ahrorov A, et al. Villin-1 and Gelsolin Regulate Changes in Actin Dynamics That Affect Cell Survival Signaling Pathways and Intestinal Inflammation. Gastroenterology. 2018;154(5):1405-20 e2.
128. Li M, Cui F, Cheng Y, Han L, Wang J, Sun D, et al. Gelsolin: role of a functional protein in mitigating radiation injury. Cell Biochem Biophys. 2015;71(1):389-96.
129. Stober VP, Lim YP, Opal S, Zhuo L, Kimata K, Garantziotis S. Inter-alpha-inhibitor Ameliorates Endothelial Inflammation in Sepsis. Lung. 2019;197(3):361-9.
130. Htwe SS, Wake H, Liu K, Teshigawara K, Stonestreet BS, Lim YP, et al. Inter-alpha inhibitor proteins maintain neutrophils in a resting state by regulating shape and reducing ROS production. Blood Adv. 2018;2(15):1923-34.
131. Barrios-Anderson A, Chen X, Nakada S, Chen R, Lim YP, Stonestreet BS. Inter-alpha Inhibitor Proteins Modulate Neuroinflammatory Biomarkers After Hypoxia-Ischemia in Neonatal Rats. J Neuropathol Exp Neurol. 2019.
132. Chromek M, Arvidsson I, Karpman D. The antimicrobial peptide cathelicidin protects mice from Escherichia coli O157:H7-mediated disease. PLoS One. 2012;7(10):e46476.

Published on 25 February 2020. Downloaded on 2/26/2020 7:17:20 AM.

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DOI: 10.1039/C9FO02998B

133. Koon HW, Shih DQ, Chen J, Bakirtzi K, Hing TC, Law I, et al. Cathelicidin signaling via the Toll-like receptor protects against colitis in mice. Gastroenterology. 2011;141(5):1852-63 e1-3.
134. Sol A, Skvirsky Y, Nashef R, Zelentsova K, Burstyn-Cohen T, Blotnick E, et al. Actin enables the antimicrobial action of LL-37 peptide in the presence of microbial proteases. J Biol Chem. 2014;289(33):22926-41.
135. Hall A, Ekiel I, Mason RW, Kasprzykowski F, Grubb A, Abrahamson M. Structural basis for different inhibitory specificities of human cystatins C and D. Biochemistry. 1998;37(12):4071-9.
136. Blotnick E, Sol A, Muhlrad A. Histones bundle F-actin filaments and affect actin structure. PLoS One. 2017;12(8):e0183760.
137. Kawasaki H, Iwamuro S. Potential roles of histones in host defense as antimicrobial agents. Infect Disord Drug Targets. 2008;8(3):195-205.
138. Lee DY, Huang CM, Nakatsuji T, Thiboutot D, Kang SA, Monestier M, et al. Histone H4 is a major component of the antimicrobial action of human sebocytes. J Invest Dermatol. 2009;129(10):2489-96.
139. Richards RC, O'Neil DB, Thibault P, Ewart KV. Histone H1: an antimicrobial protein of Atlantic salmon (Salmo salar). Biochem Biophys Res Commun. 2001;284(3):549-55.
140. Kanai Y. The role of non-chromosomal histones in the host defense system. Microbiol Immunol. 2003;47(8):553-6.
141. Sol A, Skvirsky Y, Blotnick E, Bachrach G, Muhlrad A. Actin and DNA Protect Histones from Degradation by Bacterial Proteases but Inhibit Their Antimicrobial Activity. Front Microbiol. 2016;7:1248.
142. Lalmanach G, Naudin C, Lecaille F, Fritz H. Kininogens: More than cysteine protease inhibitors and kinin precursors. Biochimie. 2010;92(11):1568-79.
143. Wang J, Wang Y, Wang J, Gao J, Tong C, Manithody C, et al. Antithrombin is protective against myocardial ischemia and reperfusion injury. J Thromb Haemost. 2013;11(6):1020-8.
144. Reiter B, Marshall VM, Philips SM. The antibiotic activity of the lactoperoxidase-thiocyanate-hydrogen peroxide system in the calf abomasum. Res Vet Sci. 1980;28(1):116-22.
145. Tenovuo J. Clinical applications of antimicrobial host proteins lactoperoxidase, lysozyme and lactoferrin in xerostomia: efficacy and safety. Oral Dis. 2002;8(1):23-9.
146. Lefkowitz DL, Lefkowitz SS, Mone J, Everse J. Peroxidase-induced enhancement of chemiluminescence by murine peritoneal macrophages. Life Sci. 1988;43(9):739-45.
147. Nakamura T, Yamagishi SI. PEDF and septic shock. Curr Mol Med. 2010;10(3):312-6.
148. Zhang SX, Wang JJ, Gao G, Shao C, Mott R, Ma JX. Pigment epithelium-derived factor (PEDF) is an endogenous antiinflammatory factor. FASEB J. 2006;20(2):323-5.
149. Westergaard UB, Andersen MH, Heegaard CW, Fedosov SN, Petersen TE. Tetranectin binds hepatocyte growth factor and tissue-type plasminogen activator. Eur J Biochem. 2003;270(8):1850-4.
150. Karmilin K, Schmitz C, Kuske M, Korschgen H, Olf M, Meyer K, et al. Mammalian plasma fetuin-B is a selective inhibitor of ovastacin and meprin metalloproteinases. Sci Rep. 2019;9(1):546.
151. Itoh H, Ide H, Ishikawa N, Nawa Y. Mast cell protease inhibitor, trypstatin, is a fragment of inter-alpha-trypsin inhibitor light chain. J Biol Chem. 1994;269(5):3818-22.
152. Schneider VA, Coorens M, Ordonez SR, Tjeerdsma-van Bokhoven JL, Posthuma G, van Dijk A, et al. Imaging the antimicrobial mechanism(s) of cathelicidin-2. Sci Rep. 2016;6:32948.
153. Lehrer RI, Ganz T. Cathelicidins: a family of endogenous antimicrobial peptides. Curr Opin Hematol. 2002;9(1):18-22.

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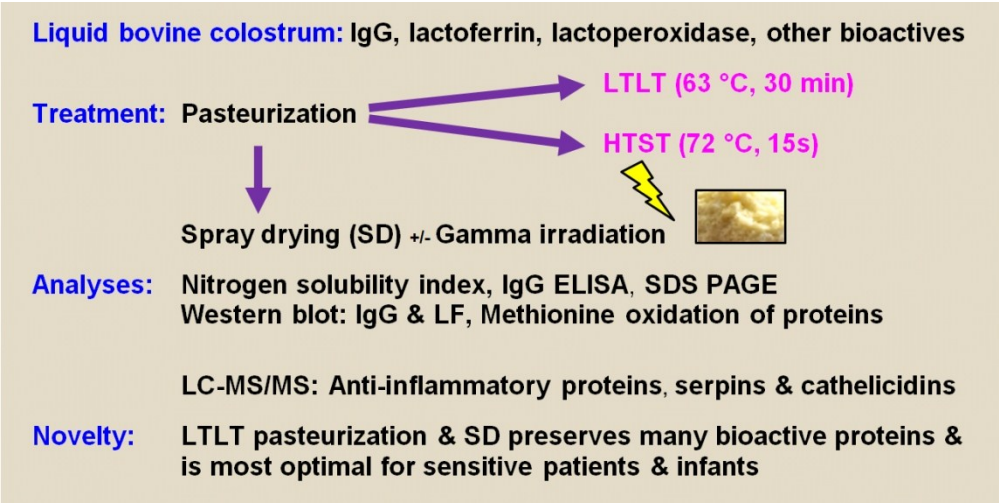

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